Transportation Research Part E 198 (2025) 104106

Contents lists available at ScienceDirect

Transportation Research Part E

journal homepage: www.elsevier.com/locate/tre

Optimizing on-demand ride-hailing services in two-sided coupled markets with impatient riders

Hui Wang [a], Yuanyuan Li [b,*], Yang Liu [c,d], Xiaowei Hu [a], Jian Wang [b]

[a] School of Transportation Science and Engineering, Harbin Institute of Technology, Harbin 150001, China
[b] School of Management, Harbin Institute of Technology, Harbin 150001, China
[c] Department of Civil and Environmental Engineering, National University of Singapore, Singapore 117576, Singapore
[d] Department of Industrial Systems Engineering and Management, National University of Singapore, Singapore 117576, Singapore

ARTICLE INFO

Keywords:
Coupled ride-hailing market
Dynamic pricing
Rider impatience
Dynamic meeting model
Reinforcement learning

ABSTRACT

This paper examines the optimal design of dynamic pricing strategies for a coupled ride-hailing market with the coexistence of ridesharing and non-ridesharing services. Riders may switch between ridesharing and non-ridesharing alternatives to maximize individual utility, which results in demand diversion. Due to the imbalance between demand and supply, riders may become impatient towards long waiting times, leading to reneging in the coupled ride-hailing systems. We first establish dynamic meeting models for ridesharing and non-ridesharing separately to characterize temporal matching patterns. Meantime, the demand diversion generated by rider switching and reneging rate resulting from rider impatience towards waiting times are integrated into the dynamic meeting models. We derive the conditions on trip fares and commission rates to ensure the existence of the coupled ride-hailing market. We then analyze the effects of the dynamic pricing strategy on the market characteristics, including waiting times and market demand. Moreover, we formulate a Markov Decision Process (MDP) model to optimize dynamic pricing strategies and the corresponding aggregated matching between demand and supply. Finally, we develop the Proximal Policy Optimization (PPO) algorithm to solve the proposed MDP model and an iteration method that uses a Lookahead algorithm to estimate the market state. Our numerical experiments reveal that higher trip fares and commission rates should be adopted under the optimal dynamic pricing strategy compared to the fixed pricing strategy. We also demonstrate the effectiveness of dynamic pricing strategy in improving system performance and matching rate. If riders are more patient with waiting times, the platform operator can improve the trip fares for ridesharing riders to better manage demand, which helps to enhance the matching rate. Meantime, both the reneging rate and the level of demand diversion can be decreased.

1. Introduction

With the development of communication technologies, mobility-on-demand services have rapidly gained a remarkable market share in urban transport systems. As an essential component of the mobility-on-demand system, ride-haling platforms provide real-time matching for riders and drivers. In practice, riders may experience high waiting times to receive a matching assignment due to the temporal or spatial supply shortage (Li et al., 2019; Bai et al., 2019). In the meantime, the impatient riders may eventually leave the system (Conolly et al., 2002; Yang et al., 2020). In this case, a ride-hailing platform may also offer ridesharing services to meet

* Corresponding author.
E-mail address: liyuanyuan@hit.edu.cn (Y. Li).

https://doi.org/10.1016/j.tre.2025.104106
Received 23 August 2024; Received in revised form 23 February 2025; Accepted 27 March 2025
Available online 12 April 2025
1366-5545/© 2025 Elsevier Ltd. All rights are reserved, including those for text and data mining, AI training, and similar technologies.

the diverse needs of riders and thus maintain the service quality, which results in a coupled ride-hailing market with ridesharing and non-ridesharing options. On the demand side, the coupled ride-hailing market is characterized by a demand coupling mechanism, allowing riders to switch between ridesharing and non-ridesharing options. On the supply side, drivers decide whether to participate in the market and which service to provide.

Integrating ridesharing with non-ridesharing services is expected to enhance the balance between demand and supply in the ride-hailing market by fully utilizing vehicle occupancy (e.g., Li and Liu, 2021). However, it is challenging to provide satisfactory matching services due to the complex interactions between demand and supply in coupled ride-hailing markets. To maximize individual utility, riders make trade-offs between ridesharing and non-ridesharing options based on travel time, trip fare, waiting time, etc. Accordingly, the demand for ridesharing may switch to non-ridesharing services, and the imbalance between demand and supply may occur. Moreover, riders may become impatient towards high waiting times and choose to renege before receiving matching service. The reneging behaviors of riders and the imbalance between demand and supply may jointly intensify the demand diversion in coupled ride-hailing markets. When ridesharing and non-ridesharing options coexist, how to quantify the impacts of rider impatience on matching patterns remains a crucial issue to address. Additionally, the matching process is executed based on the dynamic arrivals of trip requests and drivers. New modeling approaches should be developed to capture the dynamic evolution and interactions between demand and supply in coupled ride-hailing markets.

Existing literature mainly focuses on operational decisions and management strategies for non-ridesharing or ridesharing services in two-sided markets. There is limited research on systematically analyzing the operational characteristics of coupled markets involving both ridesharing and non-ridesharing. Additionally, there is a lack of consideration of the impacts of rider impatience due to waiting time on the market dynamics. To maintain the sustainable development of on-demand ride-hailing services, it is essential to account for the impacts of rider impatience on matching decisions and provide new modeling approaches to formulate the temporal evolution of matching patterns in the coupled ride-hailing market.

In this work, we develop a coupled ride-hailing market where both ridesharing and non-ridesharing services coexist. We incorporate the impacts of waiting time and rider impatience into the dynamic matching process for the coupled ride-hailing market. Specifically, the riders may become impatient and leave the ride-hailing market when the perceived waiting time increases. The impatient riders may also switch between ridesharing and non-ridesharing options to improve individual utility. Riders who switch between ride-hailing options, such as switching from ridesharing to non-ridesharing services or vice versa, can enter the respective matching pool and wait for a match. This results in a diversion of travel demand between ridesharing and non-ridesharing. By considering riders' reneging behaviors and demand diversion, we aim to model the demand-supply interaction in such ride-hailing systems and explore the optimal pricing schemes. The contributions are summarized as follows.

First, we establish a dynamic meeting model for coupled ride-hailing markets with the coexistence of ridesharing and non-ridesharing options. To describe the reneging behaviors of impatient riders, we introduce patience factors and develop an exponential distribution function to quantify the reneging probability of riders. We develop the perceived average utility function to determine the proportion of potential drivers who choose to enter the market. Among drivers entering the market, we adopt the Logit model to calculate the probability of drivers choosing each service options. Furthermore, we extend the traditional meeting function to formulate the mutual interaction between ridesharing and non-ridesharing options by capturing rider impatience. The proposed dynamic meeting model allows us to further investigate how to optimize the dynamic matching decisions and the pricing strategies in coupled ride-hailing markets with impatient riders.

Second, we analytically examine the influences of pricing strategies on market existence and demand. We derive the conditions on trip fares and commission rates to ensure the existence of the coupled ride-hailing market. Moreover, we investigate the impacts of commission rates and trip fares on demand for two ride-hailing alternatives in the coupled market. Changes in demand depend on the relationship between the waiting times of riders and the pricing scheme. It is possible to attract riders to the coupled ride-hailing market if the waiting times decrease with the commission rate or trip fare.

Third, we propose a dynamic matching optimization model to determine the time-varying matching patterns and the optimal ride-hailing pricing scheme. To maintain the sustainable development of the couple ride-hailing market, the platform should consider the impacts of matching services on social welfare while optimizing the total profit. Thus, we introduce the net economic benefits for the coupled ride-hailing market by combining the profit measure with the system-wide waiting time costs. We develop a solution method based on a reinforcement learning approach to solve the proposed optimization model. Specifically, we develop the Proximal Policy Optimization (PPO) algorithm to ensure stability and reliability in updating policy compared to traditional policy gradient methods. Moreover, we propose an iteration-based method to estimate the market states associated with both the demand and supply sides, which provides the essential inputs for implementing PPO.

Finally, we conduct extensive numerical experiments to explore the optimal design of pricing strategies and the resulting effects on system performance, using historical trip data collected in Chicago. We mainly consider two types of pricing strategies including a dynamic pricing strategy and the fixed pricing strategy. With the fixed pricing strategy, the platform charges a flat rate of trip fares from riders and commissions from drivers. And we obtain the optimal fixed pricing strategy that maximizes the net economic benefit, which serves as a benchmark for further investigation of the optimal dynamic pricing strategy. After that, we solve the proposed dynamic matching problem formulated as MDP model using the developed algorithms. We find both the trip fares and commission rate under the optimal dynamic pricing strategy are increased compared with the case under fixed pricing strategy. The dynamic pricing strategy allows the platform to balance the demand and supply in a flexible way, ensuring the improvement in net economic benefit and overall matching rate. We also find that higher rider patience leads to lower reneging rates and demand diversion in the coupled ride-hailing market. Moreover, patient ridesharing riders are inclined to wait, and impatient ridesharing riders tend to switch to using non-ridesharing.

The rest of this paper is organized as follows. Section 2 reviews relevant literature on dynamic matching and pricing problems for on-demand ride-hailing services. In Section 3, we develop meeting models for on-demand ride-haling practiced in a coupled market with both ridesharing and non-ridesharing options by incorporating rider impatience. We formulate the dynamic matching between drivers and impatient riders in the coupled ride-hailing market as a Markov Decision Process (MDP) problem in Section 4. And then, we design the solution methods to solve the proposed MDP problem. Section 5 presents our computational results. Section 6 concludes this study with our main findings.

2. Literature review

The matching optimization problem has been extensively studied to improve the balance between demand and supply for on-demand ride-hailing systems. The dynamic matching process is formulated in the two-sided markets by incorporating full-time and part-time drivers (Shi et al., 2023), vehicle rebalancing (Guo et al., 2021), joint order dispatching and driver repositioning (Sun et al., 2024), delayed matching strategies (Qin et al., 2021a), etc. Building upon two-sided matching markets, dynamic pricing strategies are designed to offer managerial insights into platform operations. For example, Chen et al. (2020) develop a dynamic meeting model to describe meeting rate over time by simultaneously capturing the macroscopic bilateral matching and the short-term market dynamics. Then, they build dynamic matching optimization problems to obtain the optimal dynamic pricing schemes that maximize platform profit or social welfare. Observing the influences of market power on pricing, Sun et al. (2019) investigate optimal pricing strategies in scenarios where the first-to-respond or closest driver is assigned to serve a trip request.

Introducing ridesharing into shared mobility systems is expected to complement the supply shortage and meet the rising travel needs of travelers. Due to the disparity in travel itineraries and time windows, riders taking ridesharing services may experience additional travel distance and trip time (Dong et al., 2018; Li et al., 2020). It is thus essential to meet the diverse travel needs of participants and maintain the service quality through pricing scheme design (Furuhata et al., 2013). In existing literature, the pricing problem for ridesharing services are explored based on user equilibrium principle (e.g., Liu and Li, 2017; Wang et al., 2019), market design theory (e.g., Zha et al., 2018; Ma et al., 2022), queueing theory (e.g., Bai et al., 2019; Nourinejad and Ramezani, 2020; Sun et al., 2022), etc. However, these studies mainly focus on optimizing matching for a single service type, i.e., ridesharing or non-ridesharing, neglecting the interaction between multiple travel options in the market.

The integration of ridesharing into the on-demand ride-hailing market leads to the coexistence of ridesharing and non-ridesharing services. From the perspective of platform operators, it is crucial to understand the interaction between two different service types and re-examine the optimal design of pricing strategies. The corresponding matching and pricing problems share similarities with the studies that focus on the competition between ride-hailing and traditional taxi services (e.g., Nie, 2017; Zhong et al., 2022; Di et al., 2022; Vignon et al., 2023), the competition between different ride-hailing platforms (e.g., Ahmadinejad et al., 2019) and the interaction between ride-hailing and public transit (e.g., Babar and Burtch, 2020). The studies above mainly focus on the competition and interaction between ride-matching services provided by different sectors. Another stream of literature investigates the complex interaction between different travel alternatives embedded in a central platform. Under such circumstances, the riders can resort to multiple different service options enabled by the same platform operator. Qin et al. (2021b) develop an integer linear program to optimize rider assignment for ride-hailing systems where riders can choose ridesharing service, non-ridesharing service, or a bundle of the two alternatives. In the presence of both ridesharing and non-ridesharing options, Zhou et al. (2023) model rider mode choice and develop pricing schemes that emphasize fairness for ridesharing participants. The riders are assumed to leave the ride-hailing market if they are not matched successfully due to long waiting times or low individual utility. Both Zhu et al. (2020) and Ke et al. (2022) focus on a stationary equilibrium problem in the coexistence of ridesharing and non-ridesharing options. Zhu et al. (2020) examine the competition between non-ridesplitting, ridesplitting and public transit in a commuting context. Ke et al. (2022) achieve the optimal operating strategies for monopoly and social optimum regimes. Unlike these studies, we model the impatience behaviors of riders due to long waiting times and the resulting reneging rate. Additionally, we explicitly capture the aggregate demand diversion between ridesharing and non-ridesharing services, considering that riders may switch between two options to maximize individual utility.

The travel behaviors of riders significantly affect the operations and management of ride-hailing markets. We mainly review the relevant literature incorporating the willingness to participate, the reneging behaviors, and order cancellation. Existing studies regarding willingness to participate concentrate on the ridesharing matching problems. Cohen et al. (2022) find that they can increase the willingness to participate of ridesharing riders by implementing appropriate compensation schemes. Exploiting the theory of planned behavior, Wang et al. (2023) show the positive effects of incentive schemes on improving the willingness to share for riders. With the incentives, the platform can retain or attract the participants and thus increase the platform revenue. However, riders entering the ride-hailing market may renege before receiving matching assignments due to long waiting times (De Vries et al., 2018). The waiting times are mainly characterized by queueing models. Considering the effects of waiting times on the utility of impatient customers, Bai et al. (2019) investigate how to balance the demand and supply via adjusting trip fares and driver wages. Economou et al. (2022) evaluate the impacts of reneging behaviors on matching decisions in two-sided markets. After the matching assignments are established, riders may also abandon the service, resulting in order cancellation (e.g., He et al., 2018; Feng et al., 2021; Xu et al., 2022; Wang et al., 2022; Li and Lo, 2022). Rider impatience and the resulted reneging behaviors have been rarely explored for coupled ride-hailing markets. In this work, we incorporate rider impatience towards waiting times into the dynamic matching problem in coupled ride-hailing markets, aiming to better understand its impacts.

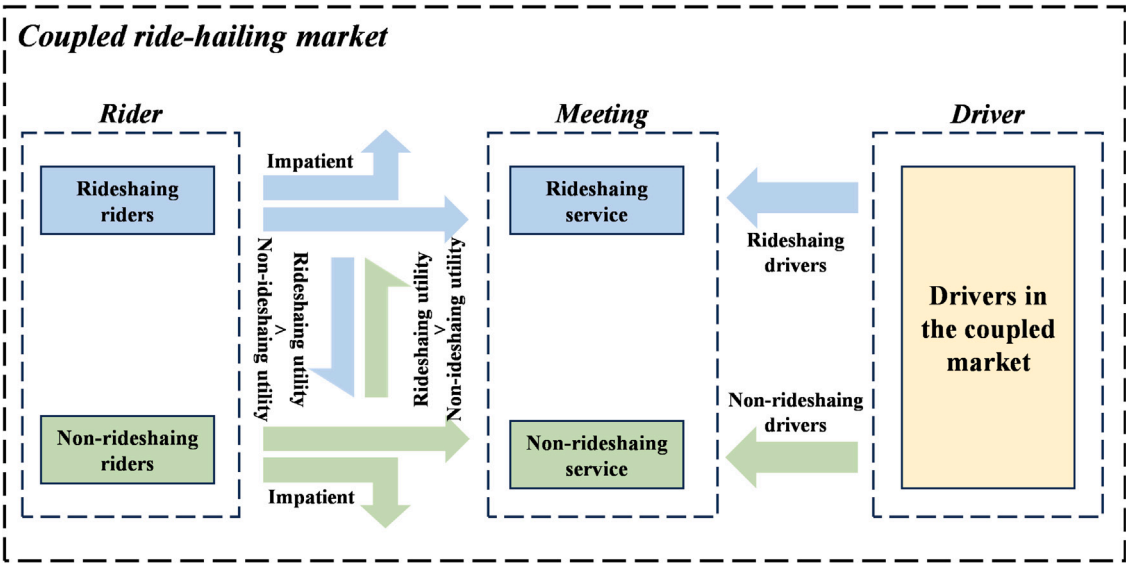


Fig. 1. Meeting process in the coupled ride-hailing market.

3. Two-sided matching in coupled ride-hailing market with impatient riders

We aim to investigate the dynamic matching problem for a coupled two-sided ride-haling market in the presence of ridesharing and non-ridesharing options. Here, we focus on introducing market coupling in on-demand ride-hailing systems in Section 3.1 and developing the meeting models in Section 3.2, respectively. We incorporate patience factors into the proposed meeting models under market coupling to characterize rider impatience. We establish the demand and supply functions in Section 3.3. Following that, we examine how the prices and commissions affect the waiting times of riders and demand for each service option in the coupled market in Section 3.4.

3.1. Description of coupled ride-hailing market

We first introduce the coupled ride-hailing market in which ride-sharing and non-ridesharing services coexist. As illustrated in Fig. 1, the demand (riders) and supply (vacant vehicles) randomly arrive at the ride-hailing platform to seek or provide ride services, respectively. Drivers in the coupled ride-hailing market decide whether to enter the system and which service option to serve. Riders specify their preferred service option as ridesharing or non-ridesharing to complete their trip requests. They have the priority in waiting for their preferred services. Before receiving services, riders can learn about the market states and the pricing strategies determined by the platform. At the same time, they may switch from their preferred service option to wait for the other option if they can maximize their individual utility. The switching between ridesharing and non-ridesharing leads to demand diversion in the coupled market. In practice, ride-hailing riders experience different waiting times depending on the market states regarding demand and supply. The riders may become impatient when the waiting time rises (Naveteur et al., 2013) and exit the system (Shortle et al., 2018). This work incorporates rider impatience and demand diversion into the two-sided matching framework.

The coupled ride-hailing market is characterized by the efficient integration of ridesharing and non-ridesharing services into a ride-hailing platform. Under the coupling mechanism, the riders can enjoy customized door-to-door matching services, and drivers may have more opportunities to increase profits. Unlike the classic two-sided market with one type of service, the riders may switch between two service alternatives, and drivers can choose the service option they want to serve after entering the market. Through pricing strategy optimization, the operator can better manage the demand and the supply for two different service options and satisfy riders' various travel needs. The enhanced vehicle capacity utilization in the coupled ride-hailing market may contribute to improved social welfare. To better regulate the ride-hailing market, it is essential to understand how the introduction of coupled ride-hailing services will affect the matching decisions of participants and system performance. However, the complicated interaction between two ride-hailing options makes it difficult to explore the resulting matching patterns. The system dynamics and rider impatience also add to the complexity of platform operations.

In this study, we aim to construct an optimization framework to investigate temporal matching strategies for on-demand ride-hailing within a unified market framework encompassing both ridesharing and non-ridesharing options. We concentrate on decision-making at the aggregate level and mainly perform macroscopic evaluation of system performance. From a methodological perspective, we can formulate matching decisions at the aggregate level by exploiting the structure of the coupled market. In what follows, we will develop modified meeting models to describe the meeting rates for ridesharing and non-ridesharing. Moreover, we define rider patience parameters and incorporate them into the proposed meeting model to account for the influences of rider impatience. Appendix summarizes all the notations used in this work.

3.2. Dynamic meeting model

We consider a coupled ride-hailing system with both ridesharing services and non-ridesharing services. Building upon the coupled ride-hailing market, we further develop the dynamic meeting models considering the impatience of riders. The driver supply and the trip requests for two service alternatives arrive continuously over time. Since the platform may execute the matching at discrete time epochs in practice, we divide the whole planning horizon into I time periods with equal length to facilitate the modeling process. Let i denote the index of time period $(t_{i-1}, t_i]$ where t_i is the corresponding decision epoch and $i \in \mathcal{I}$ ($\mathcal{I} = \{1, 2, \dots, I\}$). At each decision epoch t_i, the number of waiting riders for ridesharing service and associated vacant vehicles (idle drivers) are represented by N_{2i}^p and N_{2i}^{vc}, respectively. Let N_{1i}^p and N_{1i}^{vc} denote the number of waiting riders and vacant vehicles for non-ridesharing at decision epoch t_i, respectively. As described in Section 3.1, the demand for one service option may shift to the other. To account for this demand diversion effect, we allow the total number of waiting riders for non-ridesharing (N_{1i}^p) and ridesharing (N_{2i}^p) to include the transferred demand from ridesharing and non-ridesharing, respectively. In the remaining part of this section, we establish the meeting models for non-ridesharing and ridesharing options sequentially.

At decision epoch t_i, we define the non-ridesharing meeting rate m_{1i} as the number of matches established between riders and vacant vehicles in the non-ridesharing market. And m_{1i} is given by the function:

$$m_{1i} = M\left(N_{1i}^p, N_{1i}^{vc}\right). \tag{1}$$

The meeting rate is monotonically increasing with the number of waiting riders and vacant vehicles. Namely, the partial derivatives of the meeting rate with respect to N_{1i}^p and N_{1i}^{vc} are both positive: $\partial M / \left(\partial N_{1i}^p\right) > 0$ and $\partial M / \left(\partial N_{1i}^{vc}\right) > 0$.

On the demand side, the total number of waiting riders for non-ridesharing N_{1i}^p is affected by the new arrivals of riders, the unmatched requests from the previous time period, and the reneging rate of riders due to impatience. Additionally, the unmatched trip requests for ridesharing services can be transferred to the non-ridesharing option in the coupled ride-hailing system. To characterize the impacts of rider impatience, we incorporate a patience factor β_1 to reflect the intensity of impatience exhibited by non-ridesharing riders towards waiting time w_{1i}^p. Given waiting time w_{1i}^p, a high patience factor suggests a decreased likelihood of riders abandoning ride-hailing services in the coupled market, i.e., $\partial p_{1i}^r / \partial \beta_1 < 0$. We further define the reneging probability of non-ridesharing riders p_{1i}^r, i.e., the probability that riders abandon the non-ridesharing ride-hailing services. Assuming that the riders renege the market following a Poisson Process, we can adopt the exponential distribution to quantify the reneging probability of non-ridesharing riders: $p_{1i}^r = e^{-\beta_1 / w_{1i}^p}$ where $p_{1i}^r \in (0, 1]$. As $\partial p_{1i}^r / \partial w_{1i}^p = \frac{\beta_1}{(w_{1i}^p)^2} e^{-\beta_1 / w_{1i}^p} > 0$, the reneging probability will increase with the waiting time w_{1i}^p, which is in line with the practice.

Let L_{1i}^p denote the reneging rate of non-ridesharing riders, i.e., the number of riders who decide to exit the market due to impatience. Given the reneging probability p_{1i}^r, L_{1i}^p is given by

$$L_{1i}^p = N_{1i}^p e^{-\beta_1 / w_{1i}^p}, \tag{2}$$

where β_1 is the patience factor for non-ridesharing riders and N_{1i}^p is the number of waiting riders.

Let A_{1i}^p denote the number of new arrivals of non-ridesharing riders who announce their trip requests during time period $(t_{i-1}, t_i]$. Let T_{1i}^p (T_{2i}^p) denote the number of riders transferred from non-ridesharing (ridesharing) service to ridesharing (non-ridesharing) service. Considering the reneging behaviors of riders, we can thus formulate the number of waiting riders for non-ridesharing services during time period i as:

$$N_{1i}^p = A_{1i}^p - T_{1i}^p + T_{2i}^p + \sum_{k=0}^{i-1} (N_{1k}^p - m_{1k} - L_{1k}^p). \tag{3}$$

On the supply side, the vacant vehicles for the non-ridesharing option N_{1i}^{vc} comprise the unmatched drivers from previous periods and the newly arrived vehicles. Let A_{1i}^{vc} represent the number of vacant vehicles entering the submarket for non-ridesharing during time period i. The total number of vacant vehicles for non-ridesharing services during time period i is given by

$$N_{1i}^{vc} = A_{1i}^{vc} + \sum_{k=0}^{i-1} (A_{1k}^{vc} - m_{1k}). \tag{4}$$

Following the structure of Cobb–Douglas production function (Varian, 1992), we formulate the meeting function for time period i as below:

$$m_{1i} = A\left(N_{1i}^p - L_{1i}^p\right)^{\sigma_p} \left(N_{1i}^{vc}\right)^{\sigma_{vc}}, \tag{5}$$

where A is a parameter influenced by factors such as region, economic conditions, demographics, etc. σ_p and σ_{vc} are the elasticities of the meeting rate with respect to the numbers of waiting riders and idle drivers, respectively.

Next, we proceed to discuss the meeting rate for ridesharing services in the coupled ride-hailing market. Let L_{2i}^p denote the reneging rate of ridesharing riders in the coupled market. Similar to Eq. (2), we can establish the reneging rate of ridesharing riders as

$$L_{2i}^p = N_{2i}^p e^{-\beta_2 / w_{2i}^p}. \tag{6}$$

Then, we have the number of waiting riders and vacant vehicles for ridesharing:

$$N_{2i}^{p} = A_{2i}^{p} + T_{1i}^{p} - T_{2i}^{p} + \sum_{k=0}^{i-1}(N_{2k}^{p} - n_{kp}m_{2k} - L_{2k}^{p}), \tag{7}$$

$$N_{2i}^{vc} = A_{2i}^{vc} + \sum_{k=0}^{i-1}(A_{2k}^{vc} - m_{2k}). \tag{8}$$

Following the meeting mechanism described in Eq. (5), we can formulate the meeting function for ridesharing as follows:

$$m_{2i} = A\left(\frac{N_{2i}^{p} - L_{2i}^{p}}{n_{ip}}\right)^{\sigma_p}\left(N_{2i}^{vc}\right)^{\sigma_{vc}}, \tag{9}$$

where n_{ip} is the vehicle occupancy, i.e., the number of trip requests per vehicle for ridesharing. For simplicity, we adopt $n_{ip} = 2$ to describe the matching rule that one ridesharing vehicle can serve two trip requests at the same time. This condition will be directly applied in later sections. Active trip requests during the same time period are considered time-compatible. This means that the riders' time window requirements can potentially be satisfied. Here the meeting rate for ridesharing corresponds to the number of vehicles matched with ridesharing riders.

3.3. Demand and supply models

As discussed in Section 3.2, the meeting rates for both ridesharing and non-ridesharing services depend on their respective demand and supply. To characterize the meeting dynamics, we now formulate the demand and supply functions for two ride-hailing alternatives in the coupled market. The ride-hailing participants experience waiting times due to the existence of search frictions over time. In this part, we first analyze the average waiting times experienced by riders and drivers. Furthermore, we define the utility functions for both the demand and supply sides by incorporating the waiting times. At last, the demand and supply functions are developed.

According to Yang et al. (2010), the expected waiting times of participants in two-sided markets can be calculated based on the inter-meeting times and the number of waiting participants. The inter-meeting times refer to the time gaps between consecutive matches. On the demand side, the inter-meeting time for non-ridesharing riders is given by $1/m_{1i}$. Thus, we obtain the expected waiting times of riders choosing non-ridesharing service as below:

$$w_{1i}^{p} = N_{1i}^{p}/m_{1i}. \tag{10}$$

For riders participating in the ridesharing service, we divide their waiting time into two components. The first component represents the average waiting time experienced by a rider while the platform establishes matches among riders. This waiting time increases with the number of requests per match (i.e., the vehicle occupancy, n_{ip}) because accommodating more riders per vehicle adds to the platform operator's operational complexity. Let w_s denote the operational time required by the platform to match one rider in a shared ride. Thus, for a shared ride with 2 requests, the total operational time is $2w_s$. The second component represents the waiting time associated with matching the paired riders and vehicles. It is assumed that paired riders can be served by the same vehicle and will experience the same waiting time. The inter-meeting time for paired ridesharing riders is given by $1/m_{2i}$. Given the vehicle occupancy $n_{ip} = 2$, we obtain the average number of rider pairs in period i as $N_{2i}^{p}/2$. Therefore, paired riders experience an average waiting time of $N_{2i}^{p}/(2m_{2i})$. In summary, the average waiting time of each ridesharing rider can be expressed as follows:

$$w_{2i}^{p} = 2w_s + \frac{N_{2i}^{p}}{2m_{2i}}. \tag{11}$$

In a similar way, we can describe the waiting times of non-ridesharing drivers and ridesharing drivers as:

$$w_{1i}^{vc} = N_{1i}^{vc}/m_{1i}, \tag{12}$$

$$w_{2i}^{vc} = N_{2i}^{vc}/m_{2i}. \tag{13}$$

In the ride-hailing market, riders make decisions on whether to enter the market based on trip utility. We define the utility for non-ridesharing riders as the difference between willingness to pay for a single trip and their actual trip costs, i.e.,

$$u_{1i}^{p} = C_{\max}^{p} - C_{1i}^{p} - w_{1i}^{p}\alpha, \tag{14}$$

where $C_{\max}^{p}$ represents the maximum price a non-ridesharing rider is willing to pay for a single trip. C_{1i}^{p} represents the trip fare of non-ridesharing riders. The last item in Eq. (14) is the waiting time cost of non-ridesharing riders, with α denoting the value of their waiting time.

The demand for rides increases with the individual utility of non-ridesharing riders, i.e., $\partial A_{1i}^{p}/\partial u_{1i}^{p} > 0$. Riders choose to make trip announcements via non-ridesharing alternative if their individual utility is positive. Following existing studies (Zha et al., 2016; Yang and Yang, 2011), we construct the demand function for the non-ridesharing option as follows:

$$A_{1i}^{p} = A_{1i}^{pp}\exp(-\frac{\theta_1^{p}}{u_{1i}^{p}}), \tag{15}$$

where A_{1i}^{pp} represents the potential demand with a preference for non-ridesharing services. θ_1^p indicates the sensitivity of trip demand for ridesharing to rider utility. Riders participate in the service only when $u_{1i}^p > 0$.

Compared with non-ridesharing, ridesharing riders may experience longer trip times or waiting times (Zhan et al., 2021). Thus, we introduce Δw_i to represent the additional time experienced by riders using ridesharing compared to non-ridesharing. The utility function for ridesharing riders is defined as:

$$u_{2i}^p = C_{\max}^p - C_{2i}^p - (w_{2i}^p + \Delta w_i)\alpha', \tag{16}$$

where C_{2i}^p represents the trip fare of ridesharing riders. α' denotes the value of waiting time for ridesharing riders. Similarly, we require $u_{2i}^p > 0$ to ensure that riders are satisfied with participating in ridesharing services within the coupled market. We define the demand function for ridesharing riders as follows:

$$A_{2i}^p = A_{2i}^{pp} \exp(-\frac{\theta_2^p}{u_{2i}^p}), \tag{17}$$

where A_{2i}^{pp} represents the potential demand with a preference for ridesharing services and θ_2^p indicates the sensitivity of trip demand to the utility of ridesharing riders.

On the supply side, drivers decide whether to join the market and which service option to choose based on tradeoffs between revenue, waiting time, and operational cost. On the one hand, a driver enters the coupled ride-hailing market when the perceived average utility across the two service options is sufficiently high. On the other hand, all drivers aim to choose the service option that can maximize their individual utility. To characterize such behaviors, we first develop the utility functions for ridesharing and non-ridesharing drivers. Second, we employ a Logit model to describe drivers' service choice behaviors and calculate the proportion of drivers choosing between the two service options. Third, we construct the perceived average utility of potential drivers who may enter the coupled ride-hailing market. Lastly, we build the aggregate supply function for the coupled market. Combining the aggregate supply and the probability of serving each service option, we propose the supply functions for ridesharing and non-ridesharing services.

Let u_{1i}^{vc} and u_{2i}^{vc} denote the utility for drivers serving for non-ridesharing and ridesharing, respectively. The operational cost of a driver for serving for non-ridesharing (ridesharing) is represented by C_{1i}^{vc} (C_{2i}^{vc}). And we assume the value of waiting time for all drivers is the same as for non-ridesharing riders. u_{1i}^{vc} and u_{2i}^{vc} are expressed as:

$$u_{1i}^{vc} = C_{1i}^p\left(1 - c_i\right) - w_{1i}^{vc}\alpha - C_{1i}^{vc}, \tag{18}$$

$$u_{2i}^{vc} = 2C_{2i}^p\left(1 - c_i\right) - w_{2i}^{vc}\alpha - C_{2i}^{vc}. \tag{19}$$

Let p_i represent the proportion of drivers choosing non-ridesharing services, i.e., the probability of a driver serving non-ridesharing riders. p_i is formulated as:

$$p_i = \frac{e^{u_{1i}^{vc}}}{e^{u_{1i}^{vc}} + e^{\gamma u_{2i}^{vc}}}, \tag{20}$$

where γ is a constant parameter associated with non-ridesharing driver utility. Accordingly, the proportion of supply for ridesharing services is expressed as $1 - p_i$. The probability of becoming a ridesharing driver and the associated driver utility jointly provide the foundation for analyzing driver behaviors and supply in the coupled ride-hailing market.

A potential driver determines whether to enter the coupled market based on the perceived average utility u_i^{vc}. By combining Eqs. (18)–(20), we obtain u_i^{vc}:

$$u_i^{vc} = p_i u_{1i}^{vc} + (1 - p_i)u_{2i}^{vc}. \tag{21}$$

After that, we can build the aggregate supply function for the coupled market based on the perceived average utility of potential drivers. Let A_i^{pvc} represent the potential driver supply for couple market during time period i. To attract drivers to serve for ride-hailing in the long run, the platform should provide satisfactory matching assignments to drivers. To characterize the satisfaction of drivers, we incorporate the minimum required utility for vacant vehicles, i.e., u_{min}^{vc}. Potential drivers choose to enter the market when the average utility exceeds the minimum acceptable utility. That is, the condition $u_i^{vc} > u_{min}^{vc}$ should be satisfied. The aggregate supply function for the coupled market is given by:

$$A_i^{vc} = A_i^{pvc} \exp(-\frac{\theta_{vc}}{u_i^{vc}}), \tag{22}$$

where θ_{vc} is the sensitivity of the total vehicle supply to the perceived average utility of drivers. The supply functions for non-ridesharing and ridesharing services are expressed as:

$$A_{1i}^{vc} = p_i A_i^{vc}, \tag{23}$$

$$A_{2i}^{vc} = (1 - p_i)A_i^{vc}. \tag{24}$$

3.4. Impacts of pricing strategies

In this part, we examine the existence of the coupled ride-hailing market and further investigate the influences of pricing strategies. The results in this section provide managerial insights on regulating the coupled ride-hailing market by appropriately adjusting the trip fares and commission rates from the perspective of the platform operator.

We explore the existence of the coupled ride-hailing market with ridesharing and non-ridesharing alternatives. As discussed in Section 3.3, the platform should ensure the minimum requirement on participant utility to attract participants on both demand and supply and thus sustain ride-hailing services in the coupled markets. We obtain the analytical existence conditions for the coupled market in Proposition 1.

Proposition 1 (*Existence of Coupled Ride-Hailing Market*). *When* $\max(\frac{C_{1o}}{C_{1i}^p}, \frac{C_{2o}}{C_{2i}^p}) < c_i < 1 - \frac{u_{\min}^{vc} + p_i w_{1i}^{vc}\alpha + p_i C_{1i}^{vc} + (1-p_i) w_{2i}^{vc}\alpha + (1-p_i) C_{2i}^{vc}}{p_i C_{1i}^p + 2(1-p_i) C_{2i}^p}$, $C_{1i}^p < C_{\max}^p - w_{1i}^p\alpha$ *and* $C_{2i}^p < C_{\max}^p - (w_{2i}^p + \Delta w_i)\alpha'$, *the coupled ride-hailing market exists.*

Proof. The coupled ride-hailing market exists when the demand and supply for ride-hailing services are non-negative. On the demand side, the riders are willing to join the coupled ride-hailing market if their utility is positive. Namely,

$$\begin{cases} u_{1i}^p > 0, \\ u_{2i}^p > 0, \end{cases} \Rightarrow \begin{cases} C_{1i}^p < C_{\max}^p - w_{1i}^p\alpha, \\ C_{2i}^p < C_{\max}^p - (w_{2i}^p + \Delta w_i)\alpha'. \end{cases}$$

On the supply side, the individual utility for drivers should be larger than the minimum required utility $u_{\min}^{vc}$. That is,

$$u_i^{vc} > u_{\min}^{vc},$$

which gives us the condition on the commission rate,

$$c_i < 1 - \frac{u_{\min}^{vc} + p_i w_{1i}^{vc}\alpha + p_i C_{1i}^{vc} + (1-p_i) w_{2i}^{vc}\alpha + (1-p_i) C_{2i}^{vc}}{p_i C_{1i}^p + 2(1-p_i) C_{2i}^p}.$$

For the perspective of platform operator, the operational cost should be lower than revenue. Thus,

$$\begin{cases} c_i C_{1i}^p > C_{1o}, \\ c_i C_{2i}^p > C_{2o}, \end{cases} \Rightarrow c_i > \max(\frac{C_{1o}}{C_{1i}^p}, \frac{C_{2o}}{C_{2i}^p}). \quad \square$$

Proposition 1 implies that the existence of the coupled ride-hailing market depends on the dynamic trip fare and commission strategies. When the coupled ride-hailing market exists, the trip fares for non-ridesharing and ridesharing services must be lower than their respective derived upper bounds. If the trip fare for non-ridesharing C_{1i}^p is larger than or equal to the upper bound $C_{\max}^p - w_{1i}^p\alpha$, no riders will adopt non-ridesharing services under market coupling. And the demand for ridesharing services becomes zero if the trip fare for ridesharing C_{2i}^p is larger than or equal to the upper bound $C_{\max}^p - (w_{2i}^p + \Delta w_i)\alpha'$. At the same time, the commission rate c_i falls into the derived range as described in Proposition 1 when the coupled ride-hailing market exists. If the commission rate is higher than or equal to the derived upper bound, no drivers would like to enter the coupled ride-hailing market, which leads to a market collapse. In other words, the platform operator should set the commission rate below the derived upper bound to guarantee the satisfaction of drivers and attract them to serve for the coupled ride-hailing platform. For the platform operator, the profitability of operating coupled ride-hailing platform should also be ensured. Thus, the commission rate should be higher than the derived lower bound. Proposition 1 provides the theoretical basis to guarantee the existence and, thereby, the stable development of coupled ride-hailing services from the perspective of the platform operator.

Furthermore, we investigate how the commission rate affects the demand in the coupled ride-hailing market. As shown in Proposition 2, we present the tendency of the demand with respect to the commission rate c_i for non-ridesharing and ridesharing services in Case 1 and Case 2, respectively.

Proposition 2 (*The Impacts of Commission Rate on Demand*).

Case 1. *If* $\frac{\partial w_{1i}^p}{\partial c_i} \geq 0$, *then* $\frac{\partial A_{1i}^p}{\partial c_i} \leq 0$; *if* $\frac{\partial w_{1i}^p}{\partial c_i} < 0$, *then* $\frac{\partial A_{1i}^p}{\partial c_i} > 0$.

Case 2. *If* $\frac{\partial w_{2i}^p}{\partial c_i} \geq 0$, *then* $\frac{\partial A_{2i}^p}{\partial c_i} \leq 0$; *if* $\frac{\partial w_{2i}^p}{\partial c_i} < 0$, *then* $\frac{\partial A_{2i}^p}{\partial c_i} > 0$.

Proof. We first show the properties of new demand for non-ridesharing with respect to the commission rate at each decision epoch t_i. By taking the first-order partial derivative of A_{1i}^p in Eq. (15) with respect to c_i, we have $\frac{\partial A_{1i}^p}{\partial c_i} = \frac{\partial A_{1i}^p}{\partial u_{1i}^p}\frac{\partial u_{1i}^p}{\partial w_{1i}^p}\frac{\partial w_{1i}^p}{\partial c_i}$. Given Eq. (14), $\frac{\partial A_{1i}^p}{\partial c_i}$ can be written as $\frac{\partial A_{1i}^p}{\partial c_i} = -A_{1i}^{pp}\exp(-\frac{\theta_1^p}{u_{1i}^p})\frac{\theta_1^p\alpha}{(u_{1i}^p)^2}\frac{\partial w_{1i}^p}{\partial c_i}$ where $A_{1i}^{pp}\exp(-\frac{\theta_1^p}{u_{1i}^p})\frac{\theta_1^p\alpha}{(u_{1i}^p)^2} > 0$. Therefore, when $\frac{\partial w_{1i}^p}{\partial c_i} \geq 0$ ($\frac{\partial w_{1i}^p}{\partial c_i} < 0$), we have $\frac{\partial A_{1i}^p}{\partial c_i} \leq 0$ ($\frac{\partial A_{1i}^p}{\partial c_i} > 0$).

Case 2 in Proposition 2 can be obtained in a similar way. $\square$

In practice, changes in commission rate c_i can directly affect the decisions of drivers and thus the balance between demand and supply over time in the coupled market, which further influences the waiting time experienced by riders (w_{1i}^p and w_{2i}^p). Specifically, drivers choose between ridesharing and non-ridesharing based on the associated individual utilities. When the commission rate rises, the individual utility for ridesharing drivers (u_{2i}^{vc}) and non-ridesharing drivers (u_{1i}^{vc}) tends to decrease, resulting in a lower driver supply. In this case, the waiting times experienced by riders may increase, i.e., $\frac{\partial w_{1i}^p}{\partial c_i} \geq 0$ or $\frac{\partial w_{2i}^p}{\partial c_i} \geq 0$. At the same time, some drivers may be attracted to serve the option that is relatively more competitive or profitable when the uniform commission rate c_i is promoted. In this case, the supply for one service option may expand and then a reduction in the corresponding rider waiting time can be observed, i.e., $\frac{\partial w_{1i}^p}{\partial c_i} < 0$ or $\frac{\partial w_{2i}^p}{\partial c_i} < 0$.

Proposition 2 indicates that the effects of the commission rate on the demand for both ridesharing and non-ridesharing services depend on how it influences the corresponding waiting times for riders. The relationship between demand and the commission rate for non-ridesharing services resembles that for ridesharing services. For non-ridesharing, if a higher commission reduces the waiting time for non-ridesharing riders, an increase in demand can be expected. Similarly, the demand for ridesharing rises if a higher commission rate results in lower waiting time for ridesharing riders. The results in Proposition 2 help to understand the underlying balancing mechanism when designing the commission strategies.

At last, we examine the influences of trip fares on the demand in the coupled ride-hailing market, as shown in Proposition 3.

Proposition 3 (*The Impacts of Trip Fares on Demand*).

Case 1. *If* $\frac{\partial w_{1i}^p}{\partial C_{1i}^p} \geq -\frac{1}{\alpha}$, *then* $\frac{\partial A_{1i}^p}{\partial C_{1i}^p} \leq 0$. *And if* $\frac{\partial w_{1i}^p}{\partial C_{1i}^p} < -\frac{1}{\alpha}$, *then* $\frac{\partial A_{1i}^p}{\partial C_{1i}^p} > 0$.

Case 2. *If* $\frac{\partial w_{2i}^p}{\partial C_{2i}^p} \geq -\frac{1}{\alpha'}$, *then* $\frac{\partial A_{2i}^p}{\partial C_{2i}^p} \leq 0$. *And if* $\frac{\partial w_{2i}^p}{\partial C_{2i}^p} < -\frac{1}{\alpha'}$, *then* $\frac{\partial A_{2i}^p}{\partial C_{2i}^p} > 0$.

Proof. First, we show the impacts of the trip fare for non-ridesharing services C_{1i}^p on the demand A_{1i}^p as described in Case 1. Combining the demand of non-ridesharing riders in Eq. (15) and the rider utility u_{1i}^p in Eq. (14), we can obtain the partial derivative of A_{1i}^p with respect to the price C_{1i}^p as

$$\frac{\partial A_{1i}^p}{\partial C_{1i}^p} = A_{1i}^{pp} \exp(-\frac{\theta_1^p}{u_{1i}^p}) \frac{\theta_1^p}{(u_{1i}^p)^2} (-1 - \alpha \frac{\partial w_{1i}^p}{\partial C_{1i}^p}),$$

where all terms on the right-hand side, except for the last one, are positive. Obviously, when $\frac{\partial w_{1i}^p}{\partial C_{1i}^p} \geq -\frac{1}{\alpha}$, the last term is non-positive. As a result, we have $\frac{\partial A_{1i}^p}{\partial C_{1i}^p} \leq 0$ ($\frac{\partial A_{1i}^p}{\partial C_{1i}^p} > 0$) when $\frac{\partial w_{1i}^p}{\partial C_{1i}^p} \geq -\frac{1}{\alpha}$ ($\frac{\partial w_{1i}^p}{\partial C_{1i}^p} < -\frac{1}{\alpha}$).

Case 2 in Proposition 3 can be obtained in a similar way. □

In practice, adjusting the trip fares can influence the decision-making of the participants in the coupled ride-hailing market. The resulting change in the relationship between demand and supply further affects the waiting time of riders and the demand levels. In Proposition 3, the condition $\frac{\partial w_{1i}^p}{\partial C_{1i}^p} < -\frac{1}{\alpha}$ corresponds to the practical situation that promoting trip fare can reduce rider waiting time and the reduction rate is relatively significant. For example, when we observe a supply shortage and the pool size of the potential drivers is sufficiently large, we can increase the trip fare to attract more drivers to join the market. In this case, the waiting time of riders may drop, i.e., $\frac{\partial w_{1i}^p}{\partial C_{1i}^p} < -\frac{1}{\alpha}$. And the condition $\frac{\partial w_{1i}^p}{\partial C_{1i}^p} \geq -\frac{1}{\alpha}$ indicates that the trip fare may not be able to shorten rider waiting time effectively in practice. For example, when we observe a supply shortage and the number of potential drivers is limited, we cannot attract new drivers to enter the market by adjusting the trip fare. The premise conditions in Case 2 of Proposition 3 can be understood in a similar way.

Proposition 3 implies the change of demand against trip fare depends on how the trip fare affects the waiting time of riders. The corresponding results for non-ridesharing and ridesharing services are elaborated in Cases 1 and 2, respectively. As stated in Case 1, the demand for non-ridesharing services (A_{1i}^p) is non-increasing with trip fare (C_{1i}^p) when the condition $\frac{\partial w_{1i}^p}{\partial C_{1i}^p} \geq -\frac{1}{\alpha}$ is satisfied. In contrast, the demand for non-ridesharing services is monotonically increasing with trip fare if $\frac{\partial w_{1i}^p}{\partial C_{1i}^p} < -\frac{1}{\alpha}$. The tendency of demand against the trip fare for ridesharing services in Case 2 resembles that for non-ridesharing services. The results show that it is possible to attract riders to join the coupled ride-hailing market and thus simultaneously improve the demand for ridesharing and non-ridesharing services by appropriately increasing the trip fares. In particular, when the effects of promoting trip fare on reducing rider waiting time are relatively significant, i.e., $\frac{\partial w_{1i}^p}{\partial C_{1i}^p} < -\frac{1}{\alpha}$, the platform can improve rider demand by increasing trip fare C_{1i}^p. Otherwise, the platform should lower C_{1i}^p to improve rider demand or keep C_{1i}^p constant. The findings in Proposition 3 guide the platform in regulating the coupled ride-hailing market by designing trip fares for ridesharing and non-ridesharing alternatives, considering the effects on waiting times.

4. Dynamic matching optimization

We aim to investigate the dynamic matching problem in ride-hailing systems with the coexistence of both ridesharing and non-ridesharing options. In such a ride-hailing system, the trip requests for two types of ride-hailing services arrive dynamically and may exit the system. In the meantime, potential drivers for ride-hailing decide whether to join the coupled market, which increases the complexity of dynamic evolution on the supply side. To cope with the complexity of a coupled ride-hailing system, we exploit the Markov Decision Process (MDP) to formulate the dynamic matching between drivers and impatient riders. With this modeling framework, we perform batch matching at discrete time periods to provide the aggregated matching patterns in the system. We build the MDP model in Section 4.1 and develop the solution approach in Section 4.2.

4.1. The MDP model formulation

In this part, we propose the MDP framework to formulate the multi-period decision-making problem in the coupled ride-hailing market.

State. Let S_i represent the state of the coupled ride-hailing market at decision epoch t_i. To fully capture the system dynamics, we characterize the state space S_i with the number of waiting riders, the number of vacant vehicles, meeting rates, and reneging rates for both ridesharing and non-ridesharing alternatives. S_i is given by:

$$S_i = \left\{i, m_{1i}, m_{2i}, N^p_{1i}, N^{vc}_{1i}, L^p_{1i}, T^p_{1i}, N^p_{2i}, N^{vc}_{2i}, L^p_{2i}, T^p_{2i}\right\}. \tag{25}$$

And the state space is composed of all possible market states over the planning horizon, i.e., $\mathcal{S} = (S_i)_{i\in\mathcal{I}}$. We can observe that all state variables in state space $\mathcal{S}$ are continuous except i ($i \in \mathcal{I}$). With a continuous state space, the MDP suffers from the curse of dimensionality. Moreover, the interaction between the state variables and the state transitions over time will be complicated due to the demand diversion and reneging behaviors of riders.

Action. To improve the system performance, we dynamically optimize the ride-hailing commissions c_i and trip fares (C^p_{1i}, C^p_{2i}) over the planning horizon $i \in \mathcal{I}$. The action at decision epoch t_i is represented as a tuple $(c_i, C^p_{1i}, C^p_{2i})$.

Transition. We utilize the pre-decision state variables to describe the system state when new information becomes available and a decision has not been made. Based on Eqs. (3), (4), (7) and (8), we can present the state transition equations:

$$\begin{cases} N^p_{1i} = N^p_{1i-1} - m_{1i-1} - L^p_{1i-1} + A^p_{1i} - T^p_{1i} + T^p_{2i}, \\ N^{vc}_{1i} = N^{vc}_{1i-1} - m_{1i-1} + A^{vc}_{1i}, \\ N^p_{2i} = N^p_{2i-1} - m_{2i-1}n^p_{2i-1} - L^p_{2i-1} + A^p_{2i} + T^p_{1i} - T^p_{2i}, \\ N^{vc}_{2i} = N^{vc}_{2i-1} - m_{2i-1} + A^{vc}_{2i}. \end{cases} \tag{26}$$

As demonstrated in Eq. (26), we determine the state S_i based on the state in the previous decision stage S_{i-1}, the new information (demand and supply), and the decisions. We take the non-ridesharing option as an example to explain the development of transition functions. Given the pre-decision state S_{i-1} and a particular action taken at decision epoch t_{i-1}, we can obtain the meeting rate m_{1i-1} and the reneging rate L^p_{1i-1}. After that, we can formulate the post-decision state of the number of waiting riders ($N^p_{1i-1} - m_{1i-1} - L^p_{1i-1}$) and the number of vacant vehicles ($N^{vc}_{1i-1} - m_{1i-1}$) for non-ridesharing, respectively. The decisions made at decision epoch t_{i-1} will affect new arrivals of drivers and riders in the next decision epoch t_i. Then, before decision epoch t_i, the platform can gather the new information of non-ridesharing demand (A^p_{1i}) and supply (A^{vc}_{1i}) during the period $(t_{i-1}, t_i]$. Additionally, riders waiting for one service option may switch to the other in the coupled ride-hailing market. We incorporate demand diversion into the state transition function for the two service options. For non-ridesharing, $T^p_{2i} - T^p_{1i} > 0$ indicates an increase in rider demand for non-ridesharing during period i. In contrast, demand for ridesharing services decreases when $T^p_{2i} - T^p_{1i} > 0$.

After completing trip assignments, drivers may re-enter the system based on the perceived average utility. Those who complete their trips and then re-enter are considered new arrivals. Consequently, the potential driver supply A^{pvc}_i tends to increase during time periods with high trip completion, suggesting a temporal correlation between driver supply and rider demand in previous time periods. Accordingly, the new driver supply for non-ridesharing and ridesharing, i.e., A^{vc}_{1i} and A^{vc}_{2i} may also depend on the number of completed trips in previous periods. To explicitly address this temporal correlation, driver time schedules and service routes should be integrated into a discrete matching framework, capturing the dynamic relationship between driver availability and rider demand across time. For analytical tractability, we simplify the model by considering the potential driver supply A^{pvc}_i as an exogenous parameter and focus on temporal matching patterns at the macroscopic level.

Reward. For-profit ride-hailing platforms provide ride-matching services in order to maximize the ridership or the resulting profit. From a societal perspective, it is desirable to motivate the platform to perform efficient matching operations or implement management policies to improve social welfare. In the coupled market where both ridesharing and non-ridesharing coexist, the search friction dynamics may significantly affect the decisions of the riders and drivers. Excessive waiting times could redirect market demand towards alternative transportation modes, such as solo driving with private vehicles and public transit. As a result, the platform cannot maintain the demand or supply at a balanced level, which affects the platform operations or harms social welfare in the long run. To this end, we are motivated to optimize the platform profit and minimize the system-wide disutility associated with waiting time costs.

Specifically, we introduce the net economic benefit from the perspective of the platform operator. We define the net economic benefit as the difference between the platform's profit and the waiting time costs of ride-hailing participants. The reward at decision

epoch t_i measures the one-period net economic benefit when the platform operator takes the action of $(c_i, C^p_{1i}, C^p_{2i})$ given the system state S_i. Let C_{2o} and C_{1o} denote the operational costs from the operator's perspective for providing ridesharing and non-ridesharing services, respectively. The reward at decision epoch t_i is formulated as:

$$\begin{aligned} R(S_i,(c_i,C^p_{1i},C^p_{2i})) =& (c_i m_{2i} C^p_{2i} + c_i m_{1i} C^p_{1i}) - (m_{2i} C_{2o} + m_{1i} C_{1o}) \\ & - [\alpha' N^p_{2i}(w^p_{2i} + \Delta w_i) + \alpha w^p_{1i} N^p_{1i}] \\ & - \alpha(N^{vc}_{2i} w^{vc}_{2i} + N^{vc}_{1i} w^{vc}_{1i}), \end{aligned} \tag{27}$$

where the first term represents the revenue from providing ride-hailing services at decision epoch t_i in the coupled market. The second term corresponds to the total operational cost. The last two terms are the total waiting time costs of riders and drivers, respectively. Incorporating this performance indicator enables the platform to strike a balance between profitability and social welfare improvement.

During the optimization process, we aim to maximize the system-wide net economic benefit π:

$$\pi = \sum_{i \in \mathcal{I}} R(S_i,(c_i,C^p_{1i},C^p_{2i})). \tag{28}$$

Constraints. In addition to the state transitions equations, we impose the following constraints when optimizing the dynamic matching in coupled ride-hailing markets.

$$1 - \frac{u^{vc}_{\min} + p_i w^{vc}_{1i}\alpha + p_i C^{vc}_{1i} + (1-p_i) w^{vc}_{2i}\alpha + (1-p_i) C^{vc}_{2i}}{p_i C^p_{1i} + 2(1-p_i) C^p_{2i}} > c_i, \qquad \forall i \in \mathcal{I}, \tag{29}$$

$$\max\left(\frac{C_{1o}}{C^p_{1i}}, \frac{C_{2o}}{C^p_{2i}}\right) < c_i, \qquad \forall i \in \mathcal{I}, \tag{30}$$

$$C^p_{\max} - w^p_{1i}\alpha > C^p_{1i}, \qquad \forall i \in \mathcal{I}, \tag{31}$$

$$C^p_{\max} - (w^p_{2i} + \Delta w_i)\alpha' > C^p_{2i}, \qquad \forall i \in \mathcal{I}, \tag{32}$$

$$\min(N^p_{1i} - L^p_{1i}, N^{vc}_{1i}) \geq m_{1i}, \qquad \forall i \in \mathcal{I}, \tag{33}$$

$$\min\left(\frac{N^p_{2i} - L^p_{2i}}{2}, N^{vc}_{2i}\right) \geq m_{2i}, \qquad \forall i \in \mathcal{I}, \tag{34}$$

$$T^p_{1i} T^p_{2i} = 0, \qquad \forall i \in \mathcal{I}, \tag{35}$$

$$T^p_{1i} + T^p_{2i} \geq 0, \qquad \forall i \in \mathcal{I}, \tag{36}$$

$$c_i, C^p_{1i}, C^p_{2i} \geq 0, \qquad \forall i \in \mathcal{I}. \tag{37}$$

According to the results in Proposition 1, Eqs. (29)–(32) impose the conditions on the existence of the coupled ride-hailing market in which the ridesharing and non-ridesharing participants coexist. That is, the commission rate should vary in the range defined in Eqs. (29)–(30). The trip fare for non-ridesharing and ridesharing services should be restricted based on Eqs. (31) and (32), respectively. If these conditions on trip fare are satisfied, the riders can perceive positive utility and thus would like to join the corresponding submarket. To ensure the existence of the coupled ride-hailing market, the platform can only adjust the commission rate or trip fares with the aforementioned constraints satisfied. Eqs. (33)–(34) state that the meeting rate at each decision epoch should not exceed the minimum of potential number of trips and the total number of vacant vehicles. Eqs. (35)–(36) jointly ensure that demand diversion during each time period is associated with only one service option at most. All the decision variables are non-negative as described in Eq. (37).

4.2. Solution approach

To handle system dynamics and facilitate the development of the solution algorithm, we model the multi-period decision-making problem as the Markov decision process in Section 4.1. The dynamic pricing decisions on commission rates and trip fares made by the platform constitute the action space. We adopt multiple state variables to explicitly capture system dynamics, which results in a complicated state space. In this case, determining the optimal commission and trip fares for coupled ride-hailing markets may entail high computational costs. To address this issue, we develop a solution method based on deep reinforcement learning algorithms. Specifically, we utilize the Proximal Policy Optimization (PPO) (Schulman et al., 2017) to find an optimal policy. Recalling the models developed in Sections 3.2 and 3.3, obtaining the values of the state variables is intractable since these variables are highly interrelated. Thus, we are motivated to estimate the market state using an iteration method. With the estimated state, the PPO can learn the state space and the actions in the coupled ride-hailing market.

Next, we develop the PPO algorithm to solve the MDP model. The solution procedures are summarized in Algorithm 1. To perform PPO, we need the hyperparameters for deep learning and the waiting times estimated based on Algorithm 2. We initialize the environment based on Eqs. (25)–(26) to obtain all possible market states $\mathcal{S}$. Given a state $S_i \in \mathcal{S}$, we initialize the policy and value networks. For the policy network, we introduce $\theta((c_i, C^p_{1i}, C^p_{2i})|S_i)$ to describe the probability distribution of taking actions $(c_i, C^p_{1i}, C^p_{2i})$ given state S_i. To develop the value network, we formulate the value function at decision epoch t_i as the discounted

cumulative reward from the current decision stage i to the end of the planning horizon, i.e.,

$$V(S_i) = \max_{(c_i, C_{1i}^p, C_{2i}^p)} \left(R(S_i, (c_i, C_{1i}^p, C_{2i}^p)) + \gamma \mathbb{E}(V_{i+1}(S_{i+1}) | S_i) \right), \tag{38}$$

where $R(S_i, (c_i, C_{1i}^p, C_{2i}^p))$ represents the one-period reward at decision stage i.

For each iteration, we set the state S_0 from the environment and then perform the learning process for the whole planning horizon. Based on the current policy $\theta((c_i, C_{1i}^p, C_{2i}^p)|S_i)$, we can obtain the actual state of the coupled ride-hailing market at decision epoch t_i by calling a Lookahead method iteratively in Line 12. During this iteration process, the dynamic meeting models established in Section 3.2 are utilized. Please note Eqs. (3), (4), (7), and (8) presented in Section 3.2 are reformulated in Eq. (26) in Section 4.1. We will explain the details of the iteration process later in this section.

With the actual state, we can thus compute the reward $R(S_i, (c_i, C_{1i}^p, C_{2i}^p))$ generated at decision epoch t_i based on Eq. (27). At the same time, the constraints in Eqs. (29)–(37) must be satisfied to ensure the feasibility of the solution. The constraints in Eqs. (36)–(37) are naturally guaranteed since we always update the values of variables (T_{1i}^p, T_{2i}^p) and action $(c_i, C_{1i}^p, C_{2i}^p)$ within the feasible region. As for the constraint $T_{1i}^p T_{2i}^p = 0$, it is checked when computing the market state in Line 12. For the remaining constraints in Eqs. (29)–(32), we adopt an infeasibility penalty function $f(S_i, (c_i, C_{1i}^p, C_{2i}^p), \boldsymbol{\phi})$ to penalize any violations. This approach has been extensively utilized (e.g., Panda et al., 2024). The revised reward function is given by

$$R'(S_i, (c_i, C_{1i}^p, C_{2i}^p)) = R(S_i, (c_i, C_{1i}^p, C_{2i}^p)) - f(S_i, (c_i, C_{1i}^p, C_{2i}^p), \boldsymbol{\phi}), \tag{39}$$

where $f(S_i, (c_i, C_{1i}^p, C_{2i}^p), \boldsymbol{\phi}) = \sum_{k=1}^{K} \phi_k \mathbb{I}_k$, with $\boldsymbol{\phi} = (\phi_1, \dots, \phi_K)$ being the penalty coefficient vector for K constraints. Here, each $\mathbb{I}_k \in \{0, 1\}$ is a binary indicator of the kth constraint violation (1 if violated and 0 otherwise). The revised reward and the next state are computed in Line 13.

Then, we compute the advantages and value targets that will be utilized in performing policy and value updating, respectively. Here we first update the value network $V(S_i)$ by replacing the reward $R(S_i, (c_i, C_{1i}^p, C_{2i}^p))$ in Eq. (38) with the revised reward $R'(S_i, (c_i, C_{1i}^p, C_{2i}^p))$. That is,

$$V(S_i) = \max_{(c_i, C_{1i}^p, C_{2i}^p)} \left(R'(S_i, (c_i, C_{1i}^p, C_{2i}^p)) + \gamma \mathbb{E}(V_{i+1}(S_{i+1}) | S_i) \right). \tag{40}$$

Generally, the advantage function is defined as the difference between the action-value function and state-value function under policy $\theta((c_i, C_{1i}^p, C_{2i}^p)|S_i)$. Here, the action-value function represents the expected reward when taking action $(c_i, C_{1i}^p, C_{2i}^p)$ at state S_i. The state-value function represents the expected cumulative reward starting from state S_i. The advantage function indicates whether a specific action performs better or worse than the expected performance under the given policy. The value target $V_{target}(S_i)$ in PPO refers to an estimate of the actual value at a specific state. We compute the value target based on the reward obtained in Line 14 and the current value network.

Algorithm 1 PPO algorithm

1: **Output:**
2: Action $(c_i, C_{1i}^p, C_{2i}^p)$ and the system-wide net economic benefit π.
3: **Input:**
4: Hyperparameters: learning rate η, discount factor γ.
5: **Initialize:**
6: Initialize the environment based on Equations (25)-(26).
7: Initialize the probability distribution of decision variables $\theta((c_i, C_{1i}^p, C_{2i}^p)|S_i)$ and value network $V(S_i)$.
8: **for** $j = 1, 2, ..., J$ **do**
9: Initialize market state S_0.
10: **for** $i = 1, 2, ..., I$ **do**
11: ***Compute the rewards and states***
12: Compute the market state S_i based on $\theta((c_i, C_{1i}^p, C_{2i}^p)|S_i)$.
13: Compute the revised reward $R'(S_i, (c_i, C_{1i}^p, C_{2i}^p))$ based on Equation (39) and the next state S_{i+1}.
14: ***Compute advantages and value targets***
15: Estimate the advantages based on different states of the value network $V(S_i)$.
16: Calculate the value targets: $V_{target}(S_i) = R'(S_i, (c_i, C_{1i}^p, C_{2i}^p)) + \gamma V(S_{i+1})$.
17: ***Update the policy network***
18: Calculate policy ratio.
19: Establish the surrogate loss function.
20: $\theta((c_i, C_{1i}^p, C_{2i}^p)|S_i) \leftarrow \theta((c_i, C_{1i}^p, C_{2i}^p)|S_i) + \eta \nabla \theta(\pi_i, (c_i, C_{1i}^p, C_{2i}^p)|S_i)$.
21: ***Update the value network***
22: Establish the value loss function $V_{loss}(S_i)$.
23: Update value network $V(S_i) \leftarrow V(S_i) - \eta \nabla V_{loss}(S_i)$.
24: **end for**
25: **end for**

Following that, we proceed to update the policy network. PPO avoids dramatic updates in the policy network to maintain stability in the optimization process. The policy ratio is thus adopted in PPO to control the deviation of the new policy from the old one. It measures how much the probability of taking action $(c_i, C^p_{1i}, C^p_{2i})$ at state S_i changes from the old policy to the new policy. To avoid large updates when implementing PPO, the policy ratio should be close to 1 and vary within a small range. Combining the policy ratio and calculated advantages, we establish the surrogate loss function and minimize it during the optimization process using the gradient descent method. We can update the policy network described in Line 20 of Algorithm 1 based on the gradient.

Finally, we update the value network in PPO. For each state, we construct the value loss function $V_{loss}(S_i)$ to measure the difference between the predicted value $V(S_i)$ based on the value network and the value target $V_{target}(S_i)$. We train the value network to minimize the value loss. Thus, we can update the value network as shown in Line 23 of Algorithm 1. The new policy and value networks will guide the decision-making and the projection into future states in the next training iteration.

To implement Algorithm 1, the state S_i at each decision epoch is needed. According to Section 3.3, the waiting time on the demand (supply) side is determined by the ratio of the number of waiting riders (vacant vehicles) to the meeting rate at each decision epoch. Essentially, the number of waiting riders, the number of vacant vehicles, the meeting rate, the demand diversion values, the reneging rate, and the waiting times are endogenous and highly interconnected, with each factor influencing the others. In the following, we introduce the procedures for obtaining the state S_i based on an iteration method in line 12 of Algorithm 1.

During the iteration process, we update the demand diversion and other state variables in S_i in Eq. (25) iteratively. First, we begin with an initial guess on the demand diversion values, i.e., T^p_{1i} and T^p_{2i}. Second, we compute the state S_i and the associated waiting times given the initial values of the demand diversion. Specifically, we develop a Lookahead method to estimate variables including m_{1i}, N^p_{1i}, N^{vc}_{1i}, L^p_{1i}, w^p_{1i} and w^{vc}_{1i} for non-ridesharing, and then follow the same procedure to estimate m_{2i}, N^p_{2i}, N^{vc}_{2i}, L^p_{2i}, w^p_{2i} and w^{vc}_{2i} for ridesharing. For simplicity, we take non-ridesharing as an example to show the estimation process in the Lookahead method, as outlined in Algorithm 2. Third, the obtained estimates of these variables will be taken as inputs for updating the demand diversion values. Finally, we check if a predetermined stopping criterion is met based on the updated demand diversion and the initial values adopted in this iteration. The iteration process terminates if the stopping criterion is satisfied. Otherwise, we use the updated demand diversion values as inputs and repeat the iteration from the first step.

In the remainder of this part, we explain how the estimation is performed using the Lookahead method presented in Algorithm 2. Let $\hat{w}^p_{1i}$ and $\hat{w}^{vc}_{1i}$ represent the estimation of waiting times for non-ridesharing riders and drivers, respectively. Then, we formulate an error function e_{1i} to measure the discrepancy between our prediction and the actual magnitude of waiting times. The corresponding error function is defined as the sum of squared differences between the estimated waiting times ($\hat{w}^p_{1i}$ and $\hat{w}^{vc}_{1i}$) and the actual waiting times (N^p_{1i}/m_{1i} and N^{vc}_{1i}/m_{1i}), i.e.,

$$e_{1i} = (\hat{w}^p_{1i} - N^p_{1i}/m_{1i})^2 + (\hat{w}^{vc}_{1i} - N^{vc}_{1i}/m_{1i})^2, \tag{41}$$

To find the best estimates of w^p_{1i} and w^{vc}_{1i} that minimize the error e_{1i}, we develop a Lookahead optimizer with Adaptive Moment Estimation (Adam) as the inner-loop optimizer. We assume that the variation in market states between consecutive decision stages is not significant. Thus, we can initialize the waiting times for non-ridesharing riders and drivers using the number of waiting riders, the number of vacant vehicles, and meeting rates obtained in the previous decision epoch t_{i-1}. As shown in Lines 10-11, we set the initialized estimates for waiting times as the fast weights for non-ridesharing. Given the fast weights, we execute the estimation iteratively using the Lookahead optimizer. The main idea of the Lookahead optimizer is to update fast weights and slow weights in turn. In each iteration, the Lookahead optimizer updates the fast weights K_2 times (Lines 13–19) in the inner loop and updates the slow weight once (Lines 20–21) in the outer loop. For the inner loop, we utilize the Adam optimizer to update the fast weights. The Adam optimizer allows for a stable approximation and can enhance the convergence efficiency (Zhang et al., 2019). When the inner loop terminates, the obtained fast weights are used to calculate the slow weights in the outer loop. Finally, we recalculate the estimation error e_{1i} by taking the slow weights as the estimations $\hat{w}^p_{1i}$ and $\hat{w}^{vc}_{1i}$, and check if the stopping criteria are met in Lines 23–30. If the error e_{1i} does not exceed the predetermined upper bound $\bar{e}_{1i}$, the slow weights are regarded as the final estimation of waiting times for non-ridesharing. And the training iteration terminates. Otherwise, we update the fast weights for the next training iteration $k_1 + 1$ until $k_1 = K_1$. When the best estimates of w^p_{1i} and w^{vc}_{1i} are obtained, the values of state variables, including N^p_{1i}, N^{vc}_{1i}, L^p_{1i}, and m_{1i} are also considered the best estimates.

As for the ridesharing services, the waiting times of riders and drivers (w^p_{2i} and w^{vc}_{2i}) can be estimated by replacing the subscript 1 with 2 in Algorithm 2.

5. Numerical experiments

We perform extensive numerical experiments by solving the proposed dynamic matching model in Section 4.1 using the solution methods developed in Section 4.2. The historical trip data from Uber and Lyft, collected in Chicago in January 2020, are utilized in our experiments. We introduce the dataset and experiment setting in Section 5.1. Instead of directly solving the dynamic matching model formulated as MDP, we first conduct numerical analysis by considering a fixed pricing scheme, i.e., both flat-rate commission rate and trip fares are adopted. Based on the fixed pricing scheme, we investigate the system performance measured by net economic benefit and market states in Section 5.2. In the meantime, we aim to obtain the optimal fixed pricing scheme with a fixed commission rate and trip fares. Then, we solve the established MDP to analyze the optimal design of the dynamic pricing scheme and the resulting impacts in Section 5.3. Next, Section 5.4 investigates the influences of rider impatience towards waiting times on the dynamic pricing scheme and matching patterns. At last, we analyze the influences of values of riders' waiting time in Section 5.5.

5.1. Experiment inputs

This section introduces the experiment settings for the dynamic matching problem in the coupled ride-hailing market. First, we generate the demand and supply from real data. Second, we set the inputs for parameters characterizing the coupled ride-hailing market and the participants. Lastly, we verify the reliability of the calibrated parameters by comparing the model-based and historical matching results.

We utilize real data from ride-hailing orders in Chicago. The dataset contains information on timestamps of pickups and drop-offs, the coordinates of pickup and drop-off locations, the trip fares for each order, and the authorizations for ridesharing. The authorization for ridesharing describes whether the corresponding vehicle would like to serve ridesharing trips. In other words, the data is collected in a ride-hailing market in which ridesharing and non-ridesharing coexist, which allows us to study the coupled ride-hailing market in this work. We consider an optimization horizon of 24 h and divide it into 48 periods of equal length. We generate the potential demand (A_{1i}^{pp}, A_{2i}^{pp}) and supply A_i^{pvc} in the coupled market according to the proportional rule. And we assume that this proportion is the same across the study area and the optimization horizon. Using the total number of matched trips and vehicles in the dataset as a baseline, we adopt a potential demand-to-matched trips ratio of 2.83 and a potential supply-to-matched vehicles ratio of 1.14, respectively.

We further set the input parameters by calibration using the historical trip data. We adopt $\alpha' = \$0.24/\text{min}$ for the value of the waiting time of ridesharing riders and $\alpha = \$0.3/\text{min}$ for other participants in the coupled market. The maximum price that a rider is willing to pay is assumed to be $C_{max}^p = \$18.5$. The additional time spent by ridesharing riders compared to non-ridesharing riders Δw_i is time-varying. Based on the aforementioned real data, we obtain the time-varying Δw_i as in Fig. 2. For the base case, we adopt $C_{1i}^p = \$15.2$ and $C_{2i}^p = \$12.0$ for each rider in a non-ridesharing and ridesharing trip, respectively. The commission rate is set as $c_i = 0.2$ for all time periods. The input values for the remaining parameters are provided in Table 1.

Algorithm 2 Lookahead optimizer

Output:
The state variables: N_{1i}^p, N_{1i}^{vc}, L_{1i}^p, and m_{1i}.
The waiting times of non-ridesharing riders and drivers: $\hat{w}_{1i}^p$, and $\hat{w}_{1i}^{vc}$.
Input:
Parameters obtained in the previous decision epoch: N_{1i-1}^p, N_{1i-1}^{vc} and m_{1i-1}.
The number of riders transferred from non-ridesharing (ridesharing) to ridesharing (non-ridesharing): T_{1i}^p (T_{2i}^p).
Slow variable weight coefficient: η.
Accuracy requirement: $\bar{e}_{1i}$.
The number of fast updates per training iteration: K_2.
The number of training iterations: K_1.
Initialize:
$w_{1i}^p_fast \leftarrow N_{1i-1}^p / m_{1i-1}$.
$w_{1i}^{vc}_fast \leftarrow N_{1i-1}^{vc} / m_{1i-1}$.
for $k_1 = 1, 2, ..., K_1$ **do**
  for $k_2 = 1, 2, ..., K_2$ **do**
    Calculate N_{1i}^p and N_{1i}^{vc} based on Equation (26).
    Calculate L_{1i}^p based on Equation (2).
    Calculate m_{1i} based on Equation (5).
    Calculate e_{1i} based on Equation (41).
    Calculate gradient: ∇e_{1i}
    Calculate the changes in waiting times: Δw_{1i}^p and Δw_{1i}^{vc}.
    $w_{1i}^p_fast \leftarrow w_{1i}^p_fast - \Delta w_{1i}^p$.
    $w_{1i}^{vc}_fast \leftarrow w_{1i}^{vc}_fast - \Delta w_{1i}^{vc}$.
  end for
  $w_{1i}^p_slow \leftarrow \eta \times w_{1i}^p_slow + (1-\eta) w_{1i}^p_fast$.
  $w_{1i}^{vc}_slow \leftarrow \eta \times w_{1i}^{vc}_slow + (1-\eta) w_{1i}^{vc}_fast$.
  Calculate error e_{1i} by substituting slow weights into Equation (41).
  if error $e_{1i} < \bar{e}_{1i}$ **then**
    $\hat{w}_{1i}^p \leftarrow w_{1i}^p_slow$
    $\hat{w}_{1i}^{vc} \leftarrow w_{1i}^{vc}_slow$
    break
  else if $k_1 = K_1$ **then**
    $\hat{w}_{1i}^p \leftarrow w_{1i}^p_slow$
    $\hat{w}_{1i}^{vc} \leftarrow w_{1i}^{vc}_slow$
  end if
end for

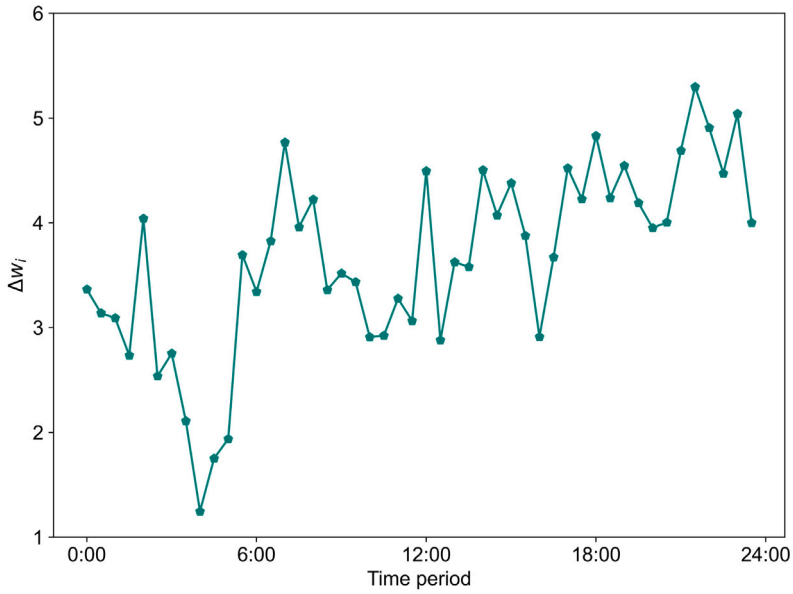


Fig. 2. Difference in travel time between ridesharing and non-ridesharing riders.

Table 1
Parameter setting.

Parameter	Value	Unit
C_{1o}	1.1	\$/trip
C_{2o}	1.4	\$/trip
C_{1i}^{vc}	4.6	\$/trip
C_{2i}^{vc}	5.7	\$/trip
u_{min}^{vc}	5	\$/trip
β_1	3	min
β_2	5	min
θ_1^p	0.5	\$
θ_2^p	0.55	\$
θ_{vc}	0.8	\$
A	0.2	–
σ_p	0.6	–
σ_{vc}	0.6	–
γ	0.15	–
$\boldsymbol{\phi}$	$(1e5, \ldots, 1e5)$	–

It is essential to ensure that the calibrated parameters produce results that comply with real-world data. Before solving the MDP model, we follow the above parameter setting and calculate the meeting rates based on the dynamic meeting model in Section 3.2. We then verify the reliability of our parameter calibration by comparing the calculated meeting rates (i.e., the model-based results) with the historical matching results. The results are shown in Fig. 3. The red solid curve and the blue dashed curve depict the meeting rate in the actual market and the calculated meeting rate, respectively. We can observe that the model-based meeting rate is close to the historical matching results. Moreover, the meeting rates during morning and evening peak hours differ significantly from off-peak periods. The above findings demonstrate that model-based matching results are in line with the matching dynamics in the real world. To assess the performance of parameter calibration, we calculate the R-squared and the mean absolute percentage error (MAPE). The R-squared is 0.98, indicating the high goodness-of-fit of the estimated meeting rates obtained based on the proposed dynamic meeting model and the calibrated parameters. The MAPE between the model-based results and the historical matching results is 2.77%, which reveals a high level of accuracy. In summary, the dynamic meeting model and the calibrated parameter inputs can fully capture the realistic characteristics of the coupled ride-hailing market.

5.2. Impacts of fixed pricing strategy

Here, we analyze the influences of fixed commission rates and trip fares on system performance measured by the net economic benefit π. At the same time, to better understand the effects of the pricing strategy, we further show how the commission rate and

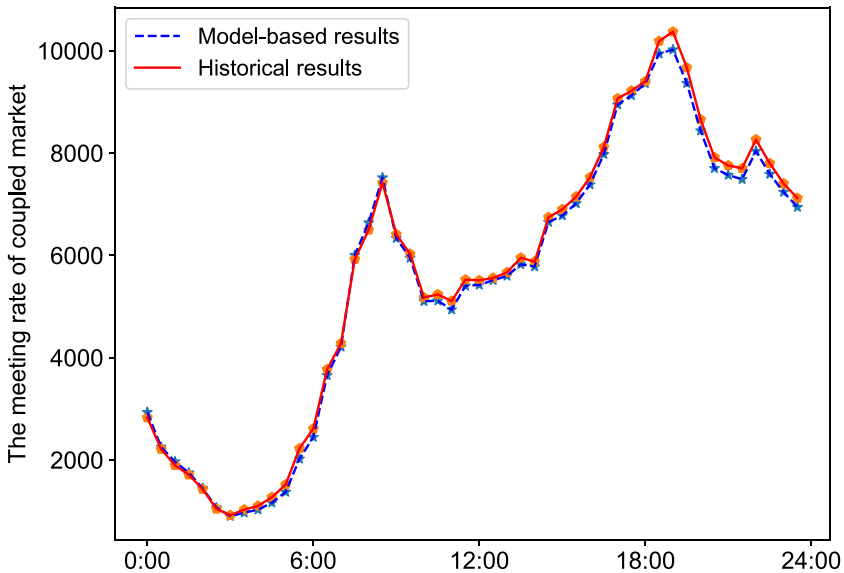


Fig. 3. The meeting rates based on the proposed dynamic meeting model and the actual market.

trip fares affect demand and the waiting times of riders. The numerical examples in this section help to validate the propositions proposed in Section 3.4.

We conduct sensitivity analysis to study the joint effects of the commission rate and trip fares on system performance. For ease of understanding, we introduce a *price multiplier* to represent the ratio of implemented trip fares to the base case where $C_{1i}^p = \$15.2$ and $C_{2i}^p = \$12.0$.

By varying the fixed commission rate in the range of $[0.10, 0.32]$ and the price multiplier in the range of $[0.80, 1.10]$, we present the net economic benefit under different combinations of commission rates and trip fares in Fig. 4. The net economic benefit π tends to increase first and then decrease as the platform increases the commission rate and trip fares. Moreover, π is non-positive in the upper left and lower right areas of Fig. 4. The non-positive π indicates that the mismatch between demand and supply is significant in the coupled ride-hailing market. Specifically, the potential drivers are more willing to enter the market when the platform adopts a low commission rate and high trip fares. In contrast, potential riders may not choose ride-hailing to complete their trips due to the high trip costs. Accordingly, an oversupply of vehicles emerges in the coupled ride-hailing market, which reduces the net economic benefit. Similarly, a high commission rate and low trip fares result in an undersupply situation and, in the meantime, low net economic benefit. This implies that the platform operator can balance the demand and supply in the coupled ride-hailing market and thus achieve optimal system performance by appropriately choosing the combined trip fares and commission rate under a fixed pricing strategy. In this example, we find the net economic benefit is maximized when $C_{1i}^p = \$16.4$, $C_{2i}^p = \$13.0$ and $c_i = 0.26$, as highlighted in the red rectangle.

We can verify the analytical results obtained in Proposition 1 based on Fig. 4. The colored area corresponds to the feasible region described in Eqs. (29)–(37). And the blank area indicates the infeasible region. The lower-left corner represents a market failure due to the violation of the platform's profitability conditions. The lower-right corner of the figure represents the infeasible region, where the minimum driver utility requirement is not satisfied.

Next, we analyze the impacts of commission rates on market demand and rider waiting times in the coupled ride-hailing market, to demonstrate the analytical results stated in Proposition 2. In this example, we focus on the time period (8:00-8:30]. The results are presented in Fig. 5. Here, the solid red curve and the black dashed curve show the partial derivative of demand and rider waiting time with respect to the commission rate, respectively. In Fig. 5(a), the demand for non-ridesharing services decreases with the commission rate c_i when $\frac{\partial w_{1i}^p}{\partial c_i} > 0$. The similar tendency can be observed for ridesharing services as depicted in Fig. 5(b). Moreover, when $\frac{\partial w_{2i}^p}{\partial c_i} < 0$, increasing commission rates promotes the demand for ridesharing services. The findings in Fig. 5 are consistent with the analytical results stated in Proposition 2.

Finally, we numerically verify the analytical findings in Proposition 3. We simultaneously show how the demand and rider waiting time change with the trip fare in Fig. 6. The results for non-ridesharing and ridesharing are provided in Figs. 6(a) and 6(b), respectively. By varying the price multiplier, we can observe the change of demand is negatively correlated with the waiting time. When $\frac{\partial w_{1i}^p}{\partial C_{1i}^p} < -\frac{1}{\alpha}$ ($\frac{\partial w_{2i}^p}{\partial C_{2i}^p} < -\frac{1}{\alpha'}$), we can find $\frac{\partial A_{1i}^p}{\partial C_{1i}^p} > 0$ ($\frac{\partial A_{2i}^p}{\partial C_{2i}^p} > 0$), which indicates the demand for non-ridesharing (ridesharing) can be improved. Otherwise, the demand is non-increasing with the trip fares.

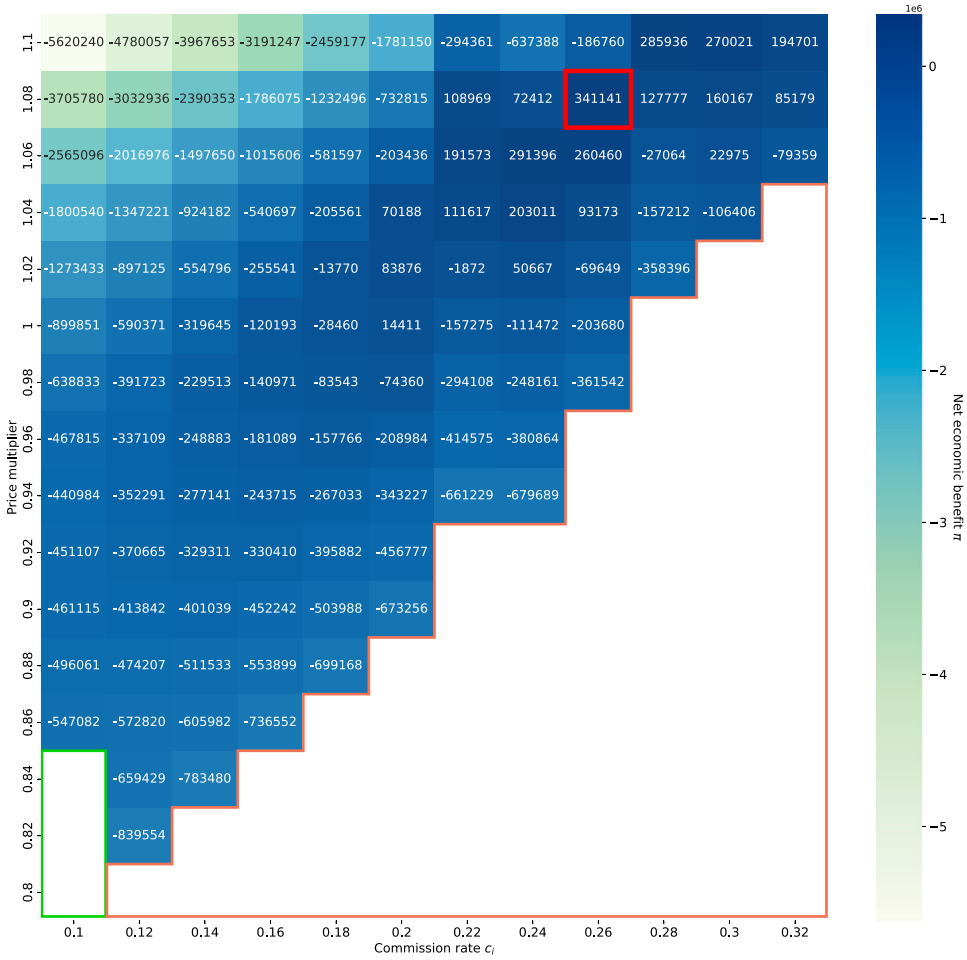


Fig. 4. The impacts of fixed commission rate and trip fares on π.

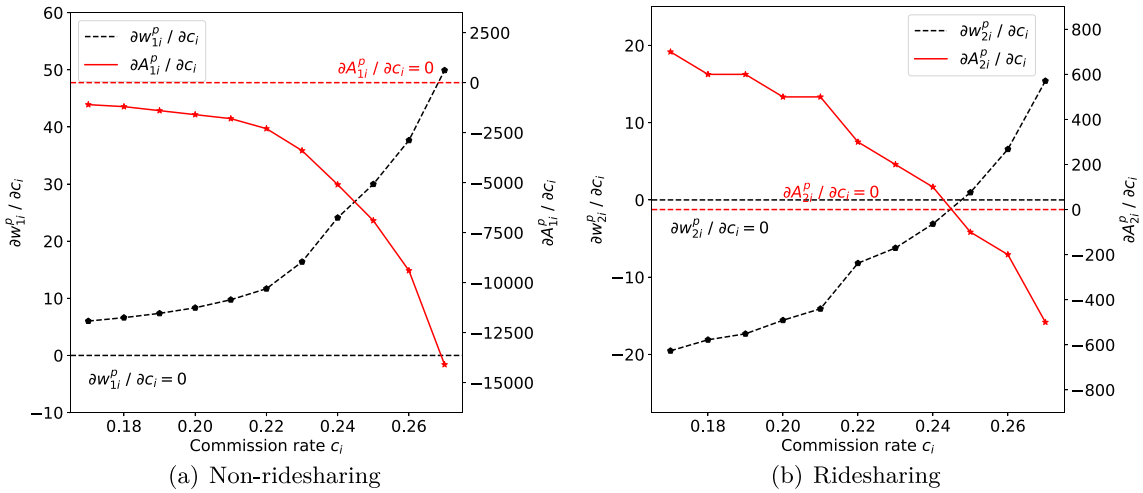
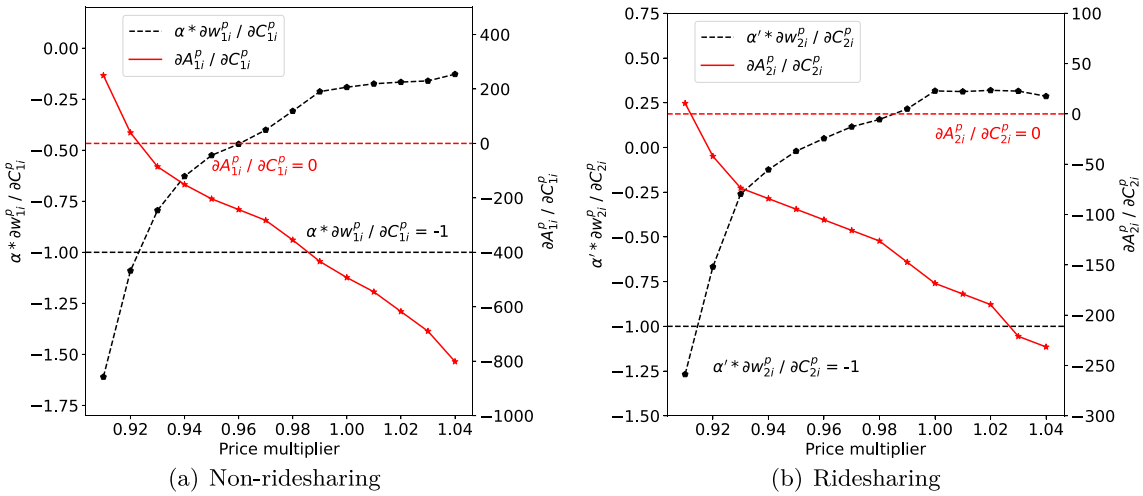
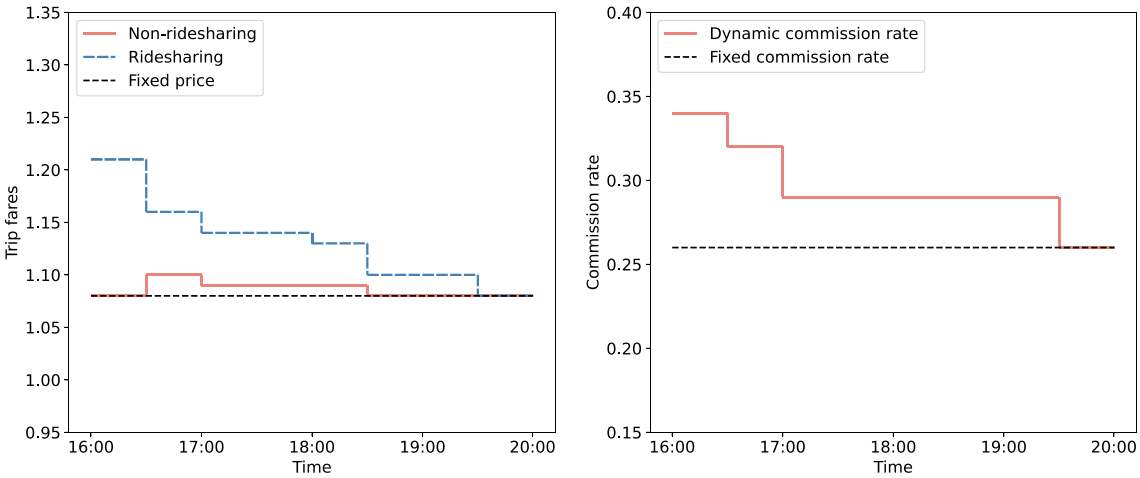


Fig. 5. The impacts of commission rates on riders' waiting times and demand.



(a) Non-ridesharing (b) Ridesharing

Fig. 6. The impacts of trip fares on riders' waiting times and demand.



(a) Trip fares C^p_{1i} and C^p_{2i} (b) Commission rate c_i

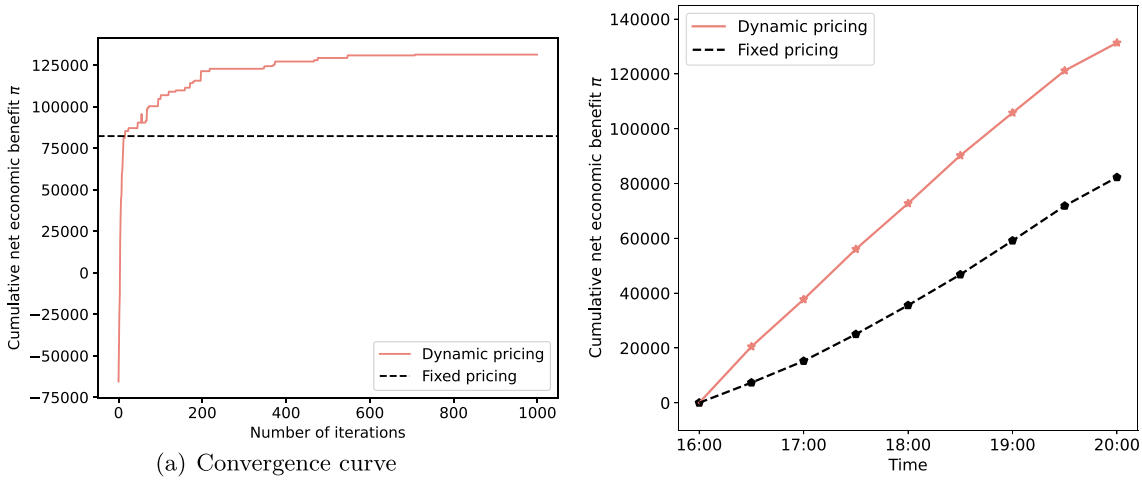
Fig. 7. Optimal pricing strategies during evening peak.

5.3. Impacts of dynamic pricing strategies

In this section, we aim to examine the optimal setting of the dynamic pricing strategy for the coupled ride-hailing system and its effectiveness in operational characteristics, including the market states and system performance. To show the effectiveness of the dynamic pricing strategies, we compare the results with that of the optimal fixed pricing strategy obtained in Section 5.2. Specifically, we set C^p_{1i} = \$16.4 and C^p_{2i} = \$13.0 and c_i = 0.26 for the fixed pricing scheme in this part. When performing the numerical experiments, we solve the proposed MDP model for the evening peak from 16:00 to 20:00 using Algorithm 1.

First, we obtain the optimal pricing strategies for each time period during the planning horizon, as shown in Fig. 7. Here, the black dashed curve depicts the fixed pricing strategy. The red solid and blue dashed curves in Fig. 7(a) represent the dynamic trip fares for non-ridesharing and ridesharing services, respectively. The red solid curve in Fig. 7(b) represents the optimal commission rate under the optimal dynamic pricing strategy. The trip fares and commission rate under the dynamic pricing scheme vary temporally over the optimization horizon. Moreover, we can observe the distinctions in trip fares between ridesharing and non-ridesharing services in Fig. 7(a). Compared with the fixed pricing scheme, the optimal fares for non-ridesharing services C^p_{1i} and ridesharing services C^p_{2i} are both higher. Meanwhile, the optimal commission rate under the dynamic pricing strategy varies around 0.29, which also increases compared to the fixed pricing strategy.

To show the effects of the optimal dynamic pricing strategy, we investigate whether we can improve system performance compared to the fixed pricing strategy. We present the system performance measured by the cumulative net economic benefits



(a) Convergence curve (b) Net economic benefit π

Fig. 8. System performance under fixed and dynamic pricing strategies.

Table 2
The demand and ridership in the coupled ride-hailing market.

Pricing strategy	Ridesharing		Non-ridesharing		Total amount	
	Demand	Ridership	Demand	Ridership	Demand	Ridership
Fixed pricing	23 172	17 448 (75.3%)	67 406	56 227 (83.4%)	90 578	73 675 (81.3%)
Dynamic pricing	15 930	14 412 (90.5%)	64 843	55 883 (86.2%)	80 773	70 295 (87.0%)

in Fig. 8. The red solid and black dashed curves show the results under dynamic and fixed pricing strategies, respectively. First, Fig. 8(a) illustrates the variation of cumulative net economic benefit (π) with the number of iterations. It shows that π can stabilize after approximately 700 iterations. Based on the tendency of π, we can numerically show the convergence of the reinforcement learning algorithm. Second, Fig. 8(b) presents the dynamic changes of π over the planning horizon. The results reveal that the cumulative net economic benefit achieved under the dynamic pricing strategy consistently outperforms that under the fixed pricing strategy. This demonstrates the effectiveness of dynamic pricing strategies in adapting to market state fluctuations.

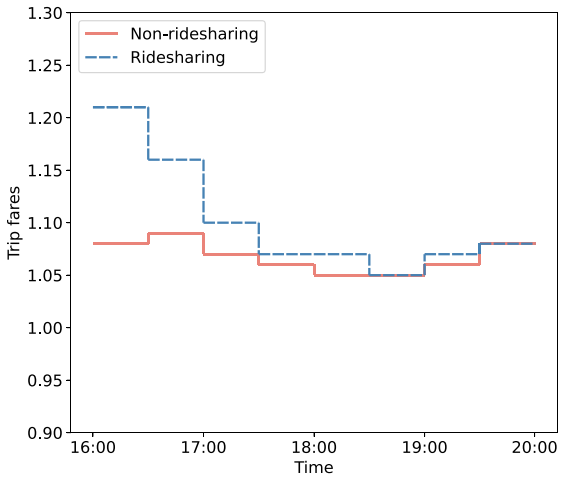
The system performance is influenced by the pricing strategy and the corresponding matching patterns of coupled ride-hailing. We further provide matching results for the coupled ride-hailing services with both ridesharing and non-ridesharing options, as given in Table 2. Specifically, we list the actual demand and the ridership for two different service options in the coupled ride-hailing market. Based on the results in Table 2, the actual demand under the dynamic pricing strategy is lower than that under the fixed pricing strategy, which is the same for both ridesharing and non-ridesharing options. The dynamic pricing strategy can lower the ridership associated with two different ride-hailing options and, thus, the overall ridership. Nevertheless, we can enhance the matching rate by adopting the dynamic pricing strategy. The relationship between demand, ridership, and matching rate reveals that dynamic pricing, as a demand management strategy, enables the platform to regulate the market and achieve a balance between supply and demand.

5.4. Analysis of rider impatience

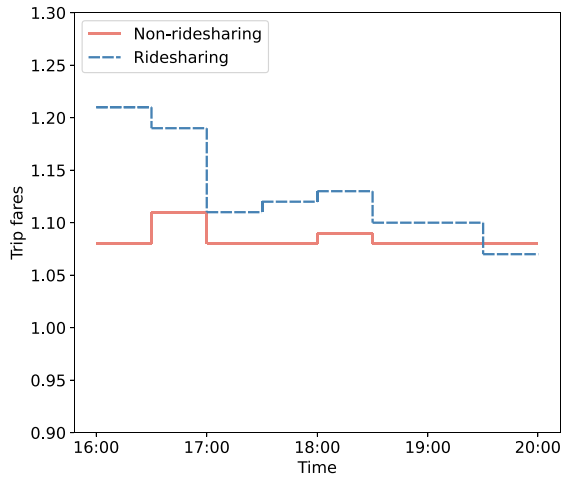
We analyze the influences of rider impatience on dynamic pricing strategy design, dynamic matching patterns, and system performance. In this part, we measure the system performance by ridership and the number of riders switching between ridesharing and non-ridesharing. By varying the patience factors (β_1 and β_2), we test the impacts of rider impatience in two scenarios: a low level of patience with $\beta_1 = 2.25$ and $\beta_2 = 3.75$ and a high level of patience with $\beta_1 = 3.75$ and $\beta_2 = 6.25$.

We obtain the optimal pricing strategy under different levels of rider impatience in Fig. 9. It shows that rider impatience affects the magnitude of optimal trip fares and commission rates across different time periods. The optimal trip fares for both non-ridesharing and ridesharing services increase slightly as riders become more patient with waiting times. The results in Fig. 9 imply that the platform should adjust the optimal pricing strategy based on rider impatience. When the riders are more patient with waiting times, the platform can improve system performance by appropriately increasing the trip fares for riders.

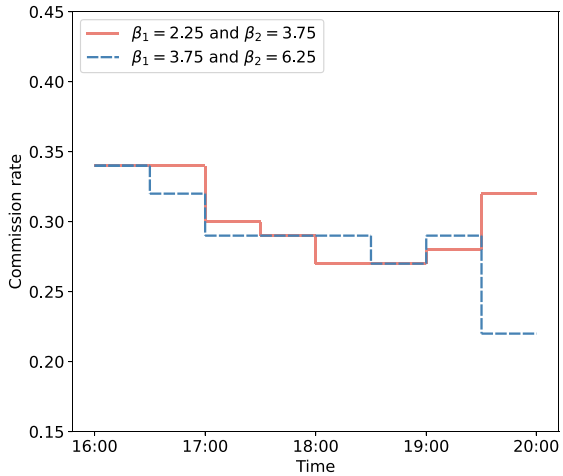
Table 3 presents the aggregated matching results and system performance. Given the same potential demand and supply, the total demand for ridesharing and non-ridesharing is lower when the riders are more patient with waiting times. We also observe that the rider matching rate (the ratio of ridership to the actual demand) for both ridesharing and non-ridesharing services has



(a) Trip fares when $\beta_1 = 2.25$ and $\beta_2 = 3.75$



(b) Trip fares when $\beta_1 = 3.75$ and $\beta_2 = 6.25$



(c) Commission rate

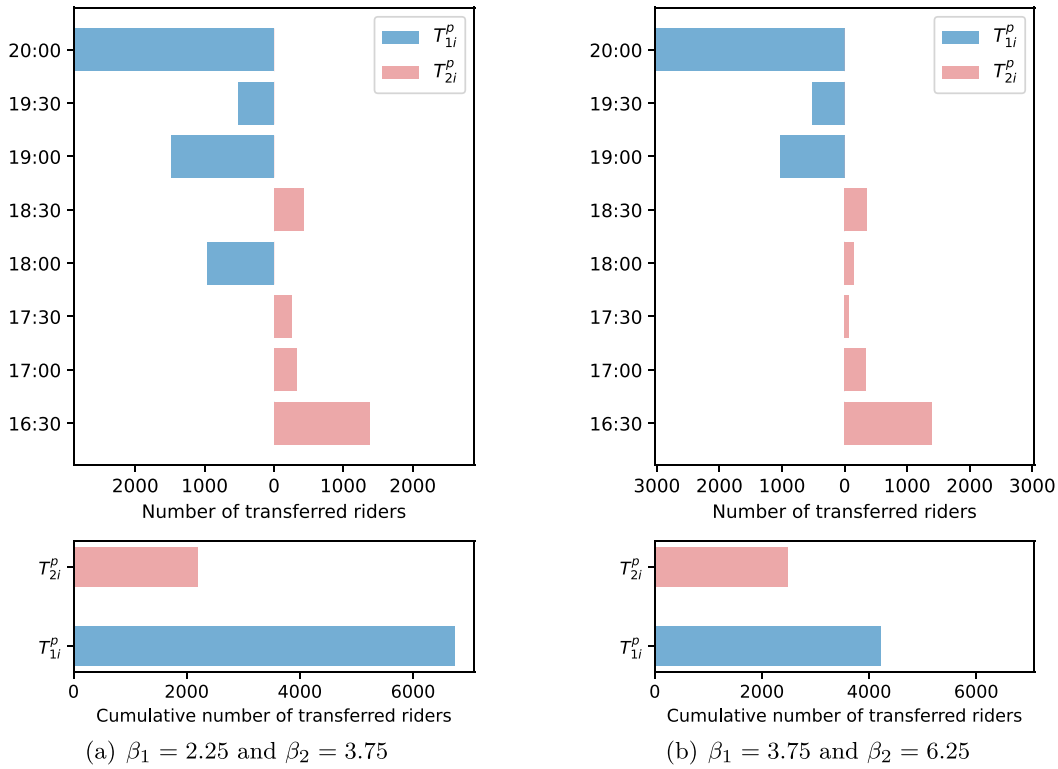
Fig. 9. Optimal pricing strategies under different levels of rider impatience.

Table 3
The influences of rider impatience on demand and matching patterns.

Patience factor	Ridesharing			Non-ridesharing		
	Demand	Ridership	Reneging	Demand	Ridership	Reneging
$\beta_1 = 2.25,\ \beta_2 = 3.75$	22 014	17 924 (81.4%)	4085	65 122	50 586 (77.7%)	10 086
$\beta_1 = 3.75,\ \beta_2 = 6.25$	19 594	17 244 (88.0%)	1177	62 938	55 360 (88.0%)	1853

increased. In our example, the matching rates for ridesharing and non-ridesharing services increase by 6.6% and 10.3%, respectively. This phenomenon indicates that the level of rider patience significantly impacts the performance of the coupled ride-hailing market. Riders with higher levels of patience are more willing to tolerate longer wait times, which significantly reduces their abandonment rates in both ridesharing and non-ridesharing services. Based on the achieved demand and matching rate, we can conclude that the dynamic pricing strategy can help efficiently manage demand when riders are more patient with waiting times.

At last, we analyze the impact of rider impatience towards waiting time on demand diversion, as shown in Fig. 10. As described in Section 3.2, a higher patience factor indicates a lower probability of reneging from the system. We further explore whether rider impatience has influenced the complicated interaction between demand and supply in the coupled ride-hailing market, leading to new patterns of demand diversion. We consider a scenario with less patient riders by setting $\beta_1 = 2.25$, $\beta_2 = 3.75$ and a scenario with more patient riders by setting $\beta_1 = 3.75$, $\beta_2 = 6.25$. Figs. 10(a) and 10(b) display the number of transferred riders between ridesharing and non-ridesharing services under these two scenarios. When we vary the patience factors, the number of riders switching service options changes dramatically over the planning horizon. Moreover, the number of riders switching from



(a) $\beta_1 = 2.25$ and $\beta_2 = 3.75$ (b) $\beta_1 = 3.75$ and $\beta_2 = 6.25$

Fig. 10. The impacts of rider impatience on demand diversion.

non-ridesharing to ridesharing decreases significantly as the patience factors increase. The results in Fig. 10 suggest that riders with higher patience towards waiting times are more likely to wait for their initially preferred service option rather than switching to an alternative, resulting in a lower demand diversion rate. This example also provides valuable insights for platform operators in managing supply to better align with demand within the coupled ride-hailing market.

5.5. Analysis of values of waiting time

In this section, we examine the influences of the values of waiting time (VOWT) of the participants in the coupled ride-hailing market. By varying the VOWT of ridesharing riders, i.e, α', we conduct the sensitivity analysis under different levels of α'/α. A higher ratio of α'/α indicates that riders perceive an larger waiting time cost if participating in ridesharing and thus have a reduced individual utility. In other words, riders using ridesharing are less tolerant with waiting time compared with non-ridesharing. In this part, we focus on the planning horizon (16:00, 20:00] with 8 time period.

Fig. 11 illustrates the trip fares for ridesharing riders (C^p_{2i}) that maximize net economic benefit under different levels of α'/α. The line chart in Fig. 11 shows the dynamic variations of ridesharing trip fares across the eight time periods, while the box plot provides an overview of the price ranges. As the ratio α'/α increases, i.e., ridesharing riders are less tolerant with waiting times, the optimal trip fare for ridesharing riders tends to decrease. The riders make tradeoffs between waiting time cost and trip fares to make their decisions on service choice, with the objective of maximizing individual utility. For riders who perceive a high waiting time cost, they may not be attracted to using ridesharing services. In this case, the platform should incentive them to enter the coupled market and choose ridesharing through lowering the trip fares. In contrast, when riders using ridesharing are more tolerant with waiting time, the platform should charge an increased trip fare from riders. The analysis here guides designing trip fares for ridesharing services according to the value of ridesharing riders' waiting time.

Following the same setting, we also explore the market performance by evaluating the number of waiting riders (N^p_{2i}) and assessing demand diversion (T^p_{1i} and T^p_{2i}) during the period (16:00, 20:00]. We present the cumulative number of waiting riders, i.e., $\sum_i N^p_{2i}$ in Fig. 12(a). Regarding the demand diversion, we calculate the proportion of ridesharing riders among the total amount of riders who switch their service options based on $\frac{\sum_i T^p_{2i}}{\sum_i (T^p_{1i}+T^p_{2i})}$. And we present the change of $\frac{\sum_i T^p_{2i}}{\sum_i (T^p_{1i}+T^p_{2i})}$ against α'/α in Fig. 12(b). As revealed, when the ratio α'/α is high, the number of waiting ridesharing riders in the market decreases rapidly, while the proportion of ridesharing riders among the total amount of riders switching service options increases significantly. Given the value of waiting time for non-ridesharing riders α, Fig. 12 shows that ridesharing riders with a relatively low value of waiting time α' are inclined to wait, whereas ridesharing riders with a high α' tend to switch to using non-ridesharing.

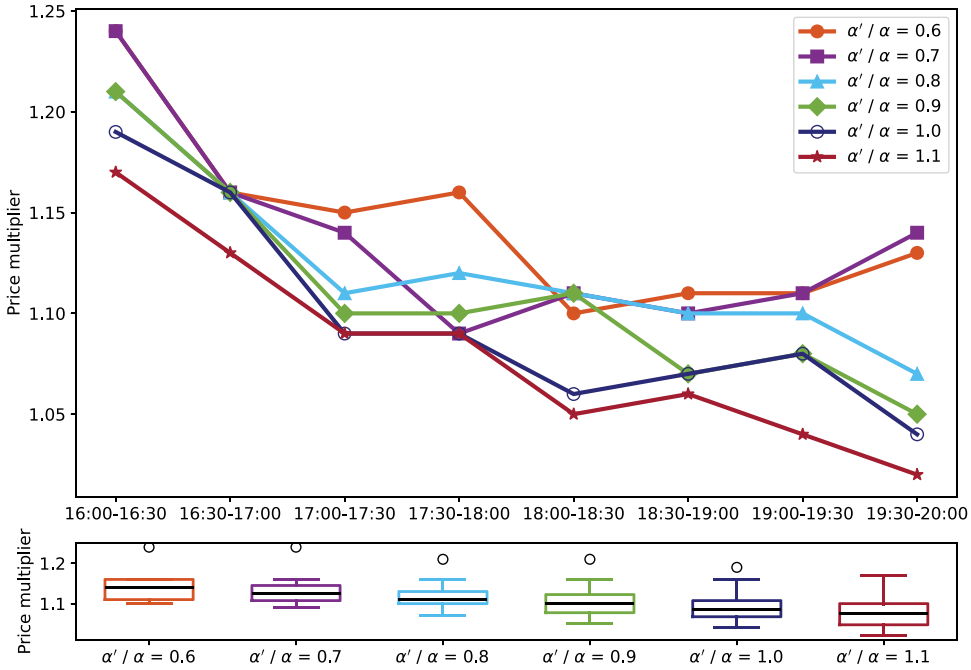


Fig. 11. Trip fares for ridesharing riders under different values of waiting time.

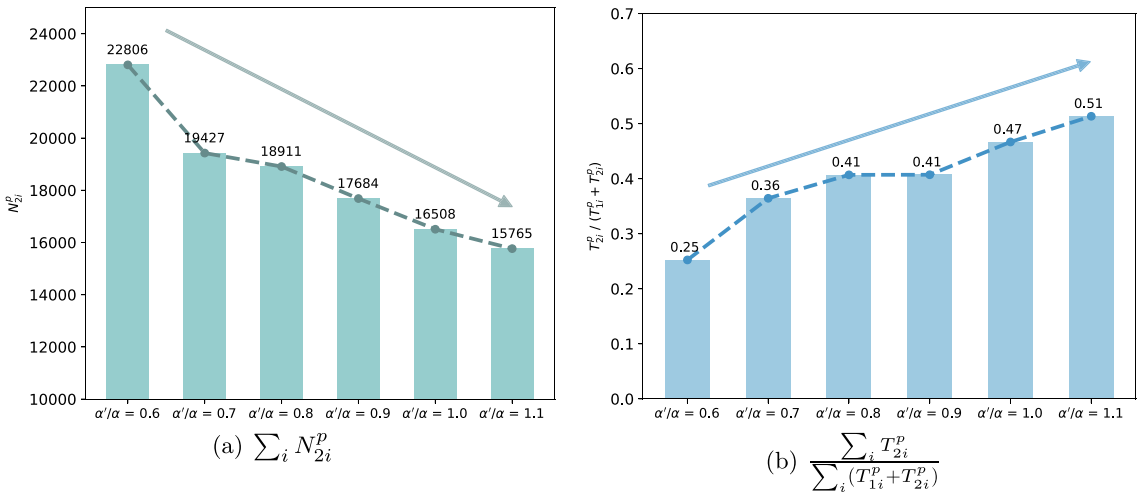


Fig. 12. Market performance under different values of waiting time.

6. Conclusions

This paper investigates the dynamic matching and pricing optimization in a coupled ride-hailing market that provides matching services via ridesharing or non-ridesharing alternatives. In practice, ride-hailing riders may become impatient with long waiting times for matching services. The riders may leave the system due to high waiting times. At the same time, riders may switch between ridesharing and non-ridesharing alternatives in that they aim to maximize individual utility, leading to demand diversion. To characterize the reneging behaviors of riders in coupled ride-hailing systems, we establish dynamic meeting models by incorporating patience factors for ridesharing and non-ridesharing riders, respectively. The proposed dynamic meeting model allows us to formulate aggregate matching decisions and capture the complicated interaction between demand and supply. Based on the meeting models, we examine the effects of pricing strategies on the existence of a coupled ride-hailing market and market characteristics, which guides the regulation of the coupled ride-hailing market from the perspective of the platform operator. Furthermore, we formulate an MDP model to explore the optimal design of combined commission and trip fare strategies to maximize net economic

Table 4
Notations.

Notation	Description
Decision variables	
c_i	The commission rate imposed at decision epoch t_i;
C_{1i}^p	Trip fare for using non-ridesharing services at decision epoch t_i;
C_{2i}^p	Trip fare for using ridesharing services at decision epoch t_i;
Subscripts or superscripts	
i	The index of time period $(t_{i-1}, t_i]$, $i \in \mathcal{I}$ and $\mathcal{I} = \{1, 2, \dots, I\}$;
t_i	Decision epoch associated with time period $(t_{i-1}, t_i]$;
Parameters	
A	The parameter involved in dynamic meeting models;
A_{1i}^{pp} (A_{2i}^{pp})	The potential demand with a preference for non-ridesharing (ridesharing) services;
A_i^{pvc}	The potential driver supply for the coupled market during time period i;
C_{max}^p	The maximum price the rider is willing to pay for a single trip;
C_{1o} (C_{2o})	The operational costs from the operator's perspective for providing non-ridesharing (ridesharing) services;
C_{1i}^{vc} (C_{2i}^{vc})	The operational cost of a driver for serving for non-ridesharing (ridesharing);
I	The total number of time periods during the planning horizon;
u_{min}^{vc}	The minimum acceptable utility for drivers;
α	Value of waiting time for all drivers and non-ridesharing riders;
α'	Value of waiting time for ridesharing riders;
w_s	The operational time required by the platform to match one rider in a shared ride;
Δw_i	The additional time experienced by riders when using ridesharing compared to non-ridesharing;
σ_p (σ_{vc})	The elasticities of the meeting rate with respect to the number of waiting riders (idle drivers);
β_1 (β_2)	Patience factor for non-ridesharing (ridesharing) riders;
θ_1^p (θ_2^p)	The sensitivity of trip demand for non-ridesharing (ridesharing) to rider utility;
θ_{vc}	The sensitivity of the potential supply to the perceived average utility of drivers;
γ	A constant parameter associated with non-ridesharing driver utility in the function p_i;
$\boldsymbol{\phi}$	The penalty coefficient vector;
Functions	
u_{1i}^p (u_{1i}^{vc})	The utility function for non-ridesharing riders (drivers) during time period i;
u_{2i}^p (u_{2i}^{vc})	The utility function for ridesharing riders (drivers) during time period i;
u_i^{vc}	The perceived average utility for drivers in the coupled ride-hailing market during time period i;
p_i	The probability of a driver participating in non-ridesharing services in the coupled ride-hailing market during time period i;
A_{1i}^p (A_{1i}^{vc})	The demand (supply) for non-ridesharing services during time period i;
A_{2i}^p (A_{2i}^{vc})	The demand (supply) for ridesharing services during time period i;
A_i^{vc}	The aggregate supply function for coupled ride-hailing market during time period i;
w_{1i}^p (w_{1i}^{vc})	The expected waiting time of non-ridesharing riders (drivers) during time period i;
w_{2i}^p (w_{2i}^{vc})	The expected waiting time of ridesharing riders (drivers) during time period i;
L_{1i}^p (L_{2i}^p)	The reneging rate of non-ridesharing (ridesharing) riders during time period i;
N_{1i}^p (N_{1i}^{vc})	The number of waiting riders (vacant vehicles) for non-ridesharing services during time period i;
N_{2i}^p (N_{2i}^{vc})	The number of waiting riders (vacant vehicles) for ridesharing service during time period i;
m_{1i} (m_{2i})	The non-ridesharing (ridesharing) meeting rate during time period i;
π	The net economic benefit.

benefit. Then, we exploit the PPO algorithm to solve the MDP model and develop an iteration method based on the Lookahead algorithm to estimate the state variables. The estimated state values are then utilized in the PPO algorithm to obtain optimal pricing decisions.

The numerical experiments are conducted using the historical trip data collected in Chicago. First, we consider a fixed pricing scheme under which the trip fares and commission rate remain constant over the entire optimization horizon. By performing sensitivity analysis, we identify the optimal fixed pricing strategy that maximizes the net economic benefit. Then, we analyze the impacts of commission rates and trip fares on market demand and rider waiting times, to validate the obtained analytical results. Second, we explore optimal decisions on dynamic pricing strategy to obtain combined trip fares and commission rate for coupled ride-hailing systems. Compared with the fixed pricing strategy, we can improve the system performance in terms of net economic benefit and matching rate. Third, we evaluate how rider impatience affects the dynamic pricing strategy, matching patterns, and system performance. Ridesharing riders should be charged higher fares when they exhibit high patience towards waiting times. Accordingly, the platform is able to reduce the reneging rate and demand diversion while increasing the matching rate. Finally, we study the impacts of values of waiting times. When the ratio of VOWT of ridesharing riders to the other participants increases, the optimal trip fare for ridesharing services and the waiting demand decrease. However, the proportion of ridesharing riders among all riders who choose to switch service options will increase. Future work may involve modeling the spatial–temporal interaction

between demand and supply in coupled ride-hailing systems. Meanwhile, we can explore more realistic pricing schemes that account for congestion conditions and trip distance. We may also extend the dynamic matching optimization problem by incorporating the heterogeneous patience levels of riders.

CRediT authorship contribution statement

Hui Wang: Writing – original draft, Visualization, Software, Methodology, Formal analysis, Data curation. **Yuanyuan Li:** Writing – review & editing, Supervision, Methodology, Funding acquisition, Conceptualization. **Yang Liu:** Writing – review & editing, Methodology. **Xiaowei Hu:** Validation, Project administration. **Jian Wang:** Validation, Supervision.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgments

The work was supported by the National Natural Science Foundation of China (Grant 72301085, 72293562), the Natural Science Foundation of Heilongjiang Province, China (Grant HSF20220088), and the Heilongjiang Postdoctoral Fund, China (Grant LBH-Z21138).

Appendix. Notations

See Table 4.

References

Ahmadinejad, A., Nazerzadeh, H., Saberi, A., Skochdopole, N., Sweeney, K., 2019. Competition in ride-hailing markets. Available at SSRN 3461119.
Babar, Y., Burtch, G., 2020. Examining the heterogeneous impact of ride-hailing services on public transit use. Inf. Syst. Res. 31 (3), 820–834.
Bai, J., So, K.C., Tang, C.S., Chen, X., Wang, H., 2019. Coordinating supply and demand on an on-demand service platform with impatient customers. Manuf. Serv. Oper. Manag. 21 (3), 556–570.
Chen, X.M., Zheng, H., Ke, J., Yang, H., 2020. Dynamic optimization strategies for on-demand ride services platform: Surge pricing, commission rate, and incentives. Transp. Res. Part B: Methodol. 138, 23–45.
Cohen, M.C., Fiszer, M.D., Kim, B.J., 2022. Frustration-based promotions: Field experiments in ride-sharing. Manag. Sci. 68 (4), 2432–2464.
Conolly, B., Parthasarathy, P., Selvaraju, N., 2002. Double-ended queues with impatience. Comput. Oper. Res. 29 (14), 2053–2072.
De Vries, J., Roy, D., De Koster, R., 2018. Worth the wait? how restaurant waiting time influences customer behavior and revenue. J. Oper. Manage. 63, 59–78.
Di, W., Tomio, M., Morikawa, T., 2022. Interrelationships between traditional taxi services and online ride-hailing: empirical evidence from xiamen, china. Sustain. Cities Soc. 83, 103924.
Dong, Y., Wang, S., Li, L., Zhang, Z., 2018. An empirical study on travel patterns of internet based ride-sharing. Transp. Res. Part C: Emerg. Technol. 86, 1–22.
Economou, A., Logothetis, D., Manou, A., 2022. The value of reneging for strategic customers in queueing systems with server vacations/failures. European J. Oper. Res. 299 (3), 960–976.
Feng, G., Kong, G., Wang, Z., 2021. We are on the way: Analysis of on-demand ride-hailing systems. Manuf. Serv. Oper. Manag. 23 (5), 1237–1256.
Furuhata, M., Dessouky, M., Ordóñez, F., Brunet, M.-E., Wang, X., Koenig, S., 2013. Ridesharing: The state-of-the-art and future directions. Transp. Res. Part B: Methodol. 57, 28–46.
Guo, X., Caros, N.S., Zhao, J., 2021. Robust matching-integrated vehicle rebalancing in ride-hailing system with uncertain demand. Transp. Res. Part B: Methodol. 150, 161–189.
He, F., Wang, X., Lin, X., Tang, X., 2018. Pricing and penalty/compensation strategies of a taxi-hailing platform. Transp. Res. Part C: Emerg. Technol. 86, 263–279.
Ke, J., Chen, X.M., Yang, H., Li, S., 2022. Coordinating supply and demand in ride-sourcing markets with pre-assigned pooling service and traffic congestion externality. Transp. Res. Part E: Logist. Transp. Rev. 166, 102887.
Li, Y., Liu, Y., 2021. Optimizing flexible one-to-two matching in ride-hailing systems with boundedly rational users. Transp. Res. Part E: Logist. Transp. Rev. 150, 102329.
Li, Y., Liu, Y., Xie, J., 2020. A path-based equilibrium model for ridesharing matching. Transp. Res. Part B: Methodol. 138, 373–405.
Li, M., Lo, H.K., 2022. Ride-sourcing service pricing and cancellation penalty under service arrival time uncertainty. Findings.
Li, W., Pu, Z., Li, Y., Ban, X.J., 2019. Characterization of ridesplitting based on observed data: A case study of Chengdu, China. Transp. Res. Part C: Emerg. Technol. 100, 330–353.
Liu, Y., Li, Y., 2017. Pricing scheme design of ridesharing program in morning commute problem. Transp. Res. Part C: Emerg. Technol. 79, 156–177.
Ma, H., Fang, F., Parkes, D.C., 2022. Spatio-temporal pricing for ridesharing platforms. Oper. Res. 70 (2), 1025–1041.
Naveteur, J., Cœugnet, S., Charron, C., Dorn, L., Anceaux, F., 2013. Impatience and time pressure: Subjective reactions of drivers in situations forcing them to stop their car in the road. Transp. Res. Part F: Traffic Psychol. Behav. 18, 58–71.
Nie, Y.M., 2017. How can the taxi industry survive the tide of ridesourcing? evidence from Shenzhen, China. Transp. Res. Part C: Emerg. Technol. 79, 242–256.
Nourinejad, M., Ramezani, M., 2020. Ride-sourcing modeling and pricing in non-equilibrium two-sided markets. Transp. Res. Part B: Methodol. 132, 340–357.
Panda, S.K., Xiang, Y., Liu, R., 2024. Dynamic resource matching in manufacturing using deep reinforcement learning. European J. Oper. Res..
Qin, G., Luo, Q., Yin, Y., Sun, J., Ye, J., 2021a. Optimizing matching time intervals for ride-hailing services using reinforcement learning. Transp. Res. Part C: Emerg. Technol. 129, 103239.
Qin, X., Yang, H., Wu, Y., Zhu, H., 2021b. Multi-party ride-matching problem in the ride-hailing market with bundled option services. Transp. Res. Part C: Emerg. Technol. 131, 103287.
Schulman, J., Wolski, F., Dhariwal, P., Radford, A., Klimov, O., 2017. Proximal policy optimization algorithms. arXiv preprint arXiv:1707.06347.
Shi, J., Li, X., Aneja, Y., Li, X., 2023. Ride-matching for the ride-hailing platform with heterogeneous drivers. Transp. Policy 136, 169–192.

Shortle, J.F., Thompson, J.M., Gross, D., Harris, C.M., 2018. Fundamentals of Queueing Theory, vol. 399, John Wiley & Sons.
Sun, J., Jin, H., Yang, Z., Su, L., 2024. Optimizing long-term efficiency and fairness in ride-hailing under budget constraint via joint order dispatching and driver repositioning. IEEE Trans. Knowl. Data Eng..
Sun, L., Teunter, R.H., Babai, M.Z., Hua, G., 2019. Optimal pricing for ride-sourcing platforms. European J. Oper. Res. 278 (3), 783–795.
Sun, Z., Xu, Q., Zhang, G., Liu, J., 2022. Pricing and matching for on-demand platform considering customer queuing and order cancellation. INFOR Inf. Syst. Oper. Res. 60 (2), 244–282.
Varian, H.R., 1992. Microeconomic Analysis, vol. 3, Norton New York.
Vignon, D., Yin, Y., Ke, J., 2023. Regulating the ride-hailing market in the age of uberization. Transp. Res. Part E: Logist. Transp. Rev. 169, 102969.
Wang, J.-P., Ban, X.J., Huang, H.-J., 2019. Dynamic ridesharing with variable-ratio charging-compensation scheme for morning commute. Transp. Res. Part B: Methodol. 122, 390–415.
Wang, L., Li, W., Weng, J., Zhang, D., Ma, W., 2023. Do low-carbon rewards incentivize people to ridesplitting? evidence from structural analysis. Transp. 50 (5), 2077–2109.
Wang, G., Zhang, H., Zhang, J., 2022. On-demand ride-matching in a spatial model with abandonment and cancellation. Oper. Res..
Xu, K., Saberi, M., Liu, W., 2022. Dynamic pricing and penalty strategies in a coupled market with ridesourcing service and taxi considering time-dependent order cancellation behaviour. Transp. Res. Part C: Emerg. Technol. 138, 103621.
Yang, H., Leung, C.W., Wong, S.C., Bell, M.G., 2010. Equilibria of bilateral taxi–customer searching and meeting on networks. Transp. Res. Part B: Methodol. 44 (8–9), 1067–1083.
Yang, H., Qin, X., Ke, J., Ye, J., 2020. Optimizing matching time interval and matching radius in on-demand ride-sourcing markets. Transp. Res. Part B: Methodol. 131, 84–105.
Yang, H., Yang, T., 2011. Equilibrium properties of taxi markets with search frictions. Transp. Res. Part B: Methodol. 45 (4), 696–713.
Zha, L., Yin, Y., Xu, Z., 2018. Geometric matching and spatial pricing in ride-sourcing markets. Transp. Res. Part C: Emerg. Technol. 92, 58–75.
Zha, L., Yin, Y., Yang, H., 2016. Economic analysis of ride-sourcing markets. Transp. Res. Part C: Emerg. Technol. 71, 249–266.
Zhan, X., Szeto, W., Shui, C., Chen, X.M., 2021. A modified artificial bee colony algorithm for the dynamic ride-hailing sharing problem. Transp. Res. Part E: Logist. Transp. Rev. 150, 102124.
Zhang, M., Lucas, J., Ba, J., Hinton, G.E., 2019. Lookahead optimizer: k steps forward, 1 step back. Adv. Neural Inf. Process. Syst. 32.
Zhong, Y., Yang, T., Cao, B., Cheng, T., 2022. On-demand ride-hailing platforms in competition with the taxi industry: Pricing strategies and government supervision. Int. J. Prod. Econ. 243, 108301.
Zhou, Z., Roncoli, C., Sipetas, C., 2023. Optimal matching for coexisting ride-hailing and ridesharing services considering pricing fairness and user choices. Transp. Res. Part C: Emerg. Technol. 156, 104326.
Zhu, Z., Qin, X., Ke, J., Zheng, Z., Yang, H., 2020. Analysis of multi-modal commute behavior with feeding and competing ridesplitting services. Transp. Res. Part A: Policy Pr. 132, 713–727.